

DATASOCIETY:

Foundations of Data and AI Literacy For Managers

Day 1



About the course

- **Instructor introduction**
- **Schedule:**
 - 4 sessions
 - 11am – 2pm EST
- **Course objectives:**
 - Define essential concepts and emerging trends in data and AI in terms of managing people, processes, and decision-making.
 - Recognize opportunities to apply data analytics to track, measure, and improve productivity and project outcomes.
 - Develop acumen to initiate, manage, or contribute to data and AI projects within your organization, in collaboration with technical resources.
 - Recognize the role of data governance in ensuring data integrity, reliability, resourcing, and compliance.
 - Discuss strategies to foster a data-driven culture among teammates.



Agenda

Day 1

- What is data literacy?
- Fundamentals of data
- Introduction to AI
- Data analytics

Day 2

- Data management
- Data governance
- Data project lifecycle

Day 3

- AI for data analytics
- AI implementation

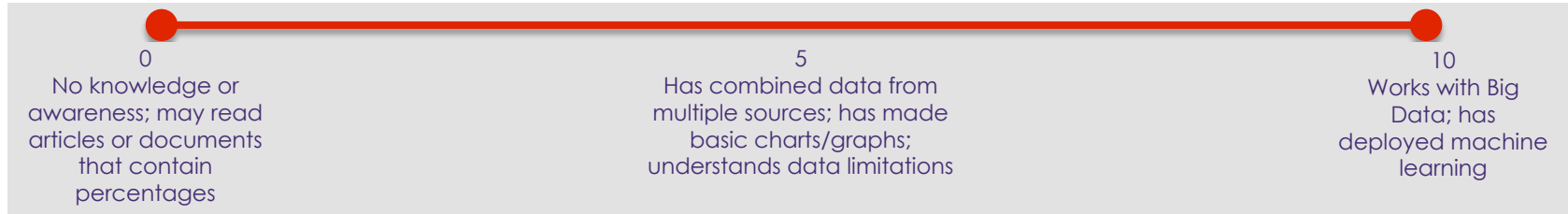
Day 4

- Teams and tools
- Building a data-driven culture
- Ethical and responsible use of data & AI
- What's next?



Polling question

What would you rate your current data literacy level on a scale of 0 -10?



Agenda

Day 1

- **What is data literacy?**
- Fundamentals of data
- Introduction to big data and AI
- Data analytics

Defining data literacy

- Data is a valuable asset for businesses, but data literacy is what brings it to life.
- **Data literacy** is the ability to:
 - **Read**
 - **Write**
 - **Communicate with data** in context — in both work and life through development of one's mindset, language and skills.



What is required to be data literate?

MINDSET

Being open, willing & curious.

Seeing the world through data glasses.

LANGUAGE

Business, Data + Analytics Acumen.

“Speaking Data” as a shared language.

SKILLS

Thinking critically.

Engaging others.

Applying data constructively.

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Developing mindset

- Cultivating a **data-driven mindset** requires a **shift** from relying on intuition and gut feelings to embracing the power of data.
- To achieve this, it is vital to:
 - Understand the data and steps performed within data analysis.
 - Approach data and analysis with a curious and questioning mind.



Enhancing skills

- Data skills is not just about using data, tools or technology.
- To truly unlock the power of data, begin by cultivating essential **non-technical** skills:
 - **Critical thinking** to ask meaningful questions to the data
 - **Collaboration skills** to work effectively with others and engaging team members, stakeholders
 - Ability to take **constructive and ethical action** when using data



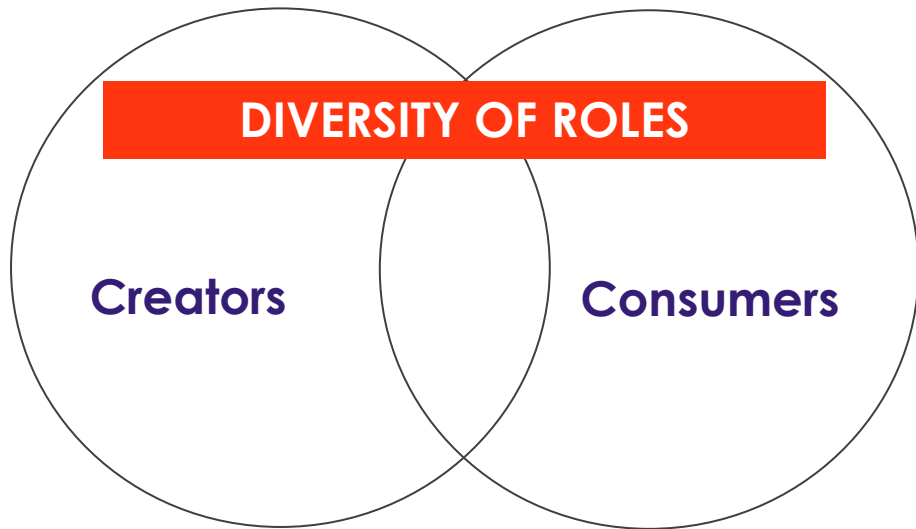
Chat question

How do you interpret the phrase
"speaking data"?



Need for a shared language

- We interact with people from diverse backgrounds:
 - Industry sector experience
 - Business domain experience
 - Data versus analytics backgrounds
 - Global vs. local
 - Veterans vs. rookies
- Diversity is desired, and healthy. Yet it can result in natural communication friction and disconnects.
- **Creating and nurturing a shared language is fundamental.**



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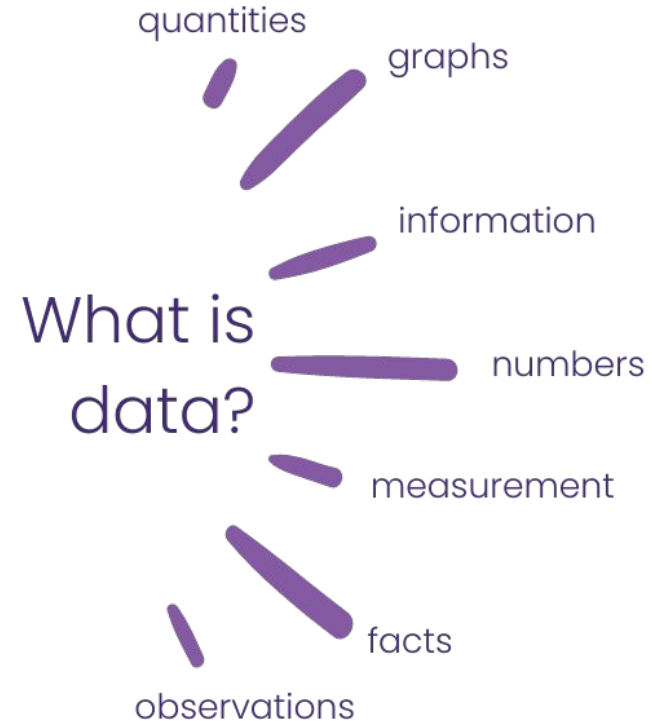
Agenda

Day 1

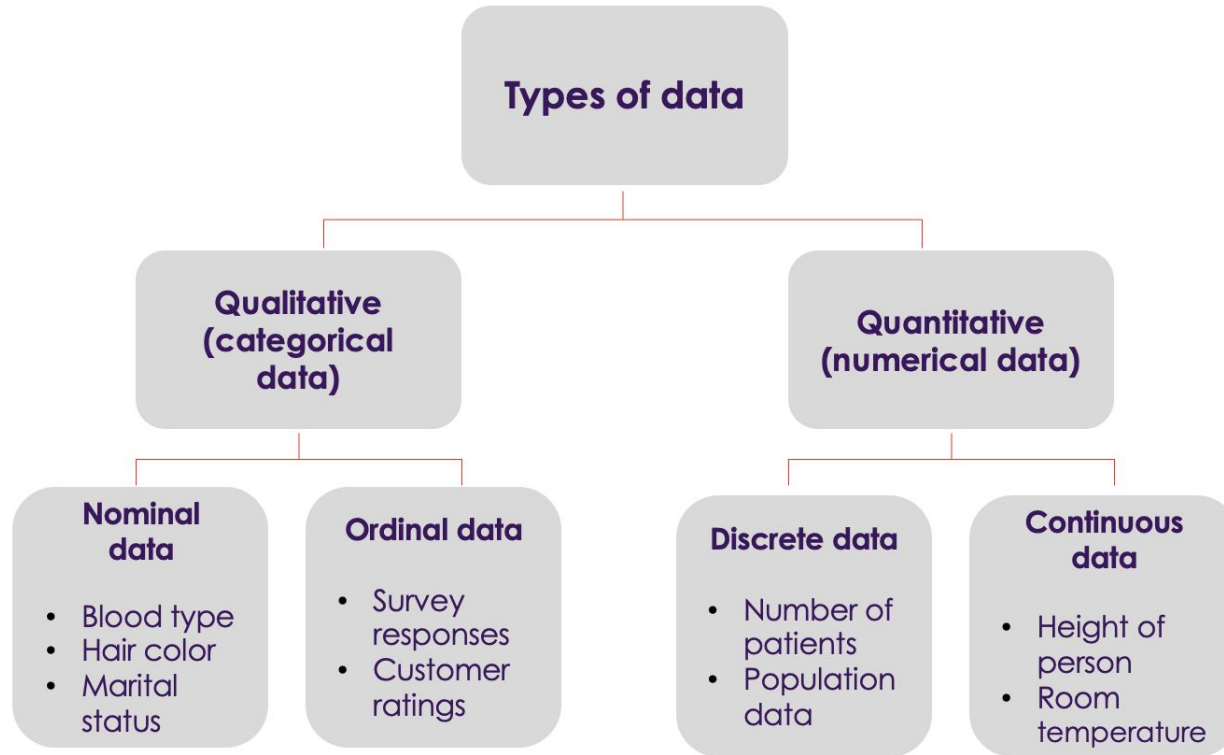
- What is data literacy?
- **Fundamentals of data**
- Introduction to big data and AI
- Data analytics

What is data?

- **Data** is everywhere. So much so that there's possibly no field that doesn't deal with data.
- It is **information, facts, or numbers** collected to be examined, considered, and used to help decision-making.
- The data universe continues to grow relentlessly.



Types of data



Data formats

Unstructured



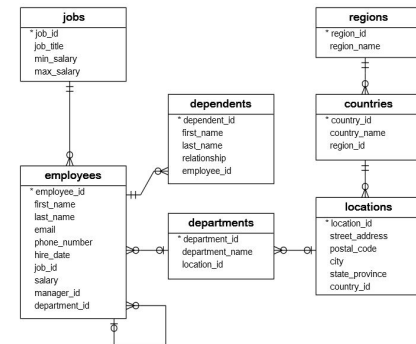
Semi/Quasi-structured

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<Book ISBN="0553212419">
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  <author>Sir Arthur Conan Doyle</author>
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  <title>Nothing Like It In the World</title>
  <author>Stephen E. Ambrose</author>
</Book>
</Books>
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Sep 17 02:33:08.536 [debug]
connection_edge_process_relay_cell(): Now seen 1802 relay cells here (command 2, stream 5845).
Sep 17 02:33:08.536 [debug]
connection_edge_process_relay_cell(): circ deliver_window now 933.

Structured

y1	x1	x2	x3





Polling question

Question #1: What data format would emails fall under?

- a. Structured data
- b. Semi-structured data
- c. Unstructured data

Question #2: What data format would research papers fall under?

- a. Structured data
- b. Semi-structured data
- c. Unstructured data

Sources of data



Internal

- Information Resources Management (IT)
- Global Talent Management systems(HR)
- Comptroller and Global Financial Services systems (Finance)
- Supply chain records
- Service records

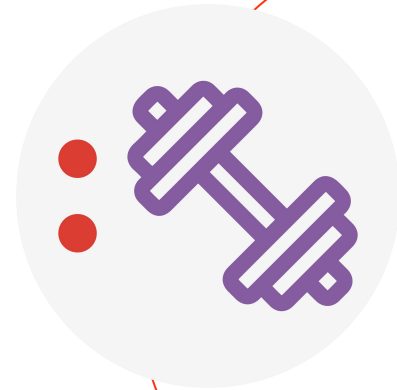


External

- Publicly-accessible APIs
 - e.g., api.data.gov
- Other open data sources
 - e.g., data.worldbank.org
- Large businesses are increasingly giving people access to their data
 - e.g., Expedia

Exercise: Data-You use it more than you think

- a. **Consider the data you use or interact with on a daily basis:**
 - Write down a list of 3-5 types of data/information you rely on regularly.
- b. **Identify the characteristics of each type of information:**
 - What type of data is it?
 - Is it structured, semi-structured or unstructured?
 - What is the source of this information?
- c. **Reflect on your relationship with data:**
 - How do you use this data to make decisions or support your choices?
 - What data sources do you trust, and why do you consider them reliable?



Refer to **page 4** in the participant guide for this activity.

Debrief time

Share your findings and responses with the whole group.



Break

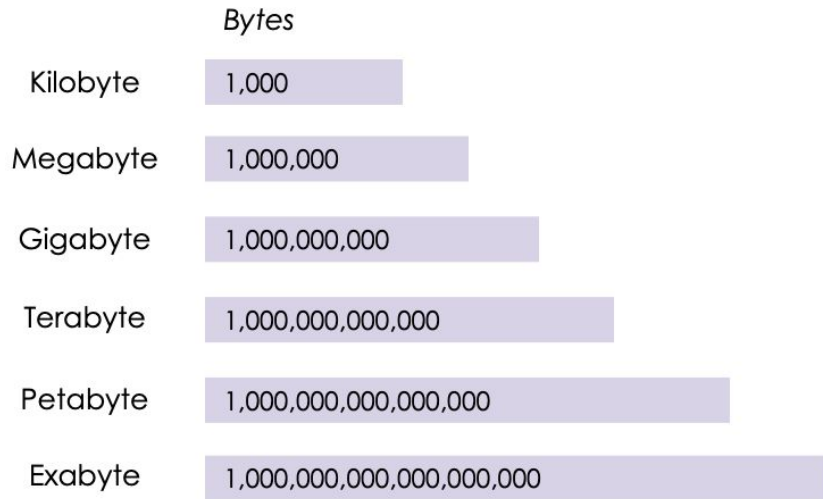


Agenda

- What is data literacy?
- Fundamentals of data
- **Introduction to big data and AI**
- Data analytics

Did you know?

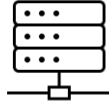
198,947,789,624,256 GB
data is generated everyday!



What is big data?

*“Big data” refers to a **large volume of data** that can be **mined for information** and **used in machine learning** projects and other analytics applications.*

Characteristics of big data



High volume

Data size is described in interabytes, petabytes, and exabytes!

High velocity

Big data flows from sources at a rapid and continuous pace



High variety

Big data comes in different formats from heterogeneous sources

Heterogeneous data

- Big data technology offers incredible opportunities for accelerating discoveries and innovations, but if not used responsibly, it poses significant risks.
- It is important to ensure there is a broad spectrum of data, i.e., different kinds of objects are represented.
- Data that reflects **a variety of formats, sources, experiences and levels of details** can help fill in the gaps and provide a more rounded and accurate picture.



Image source: [Voneff at The Noun Project](#)

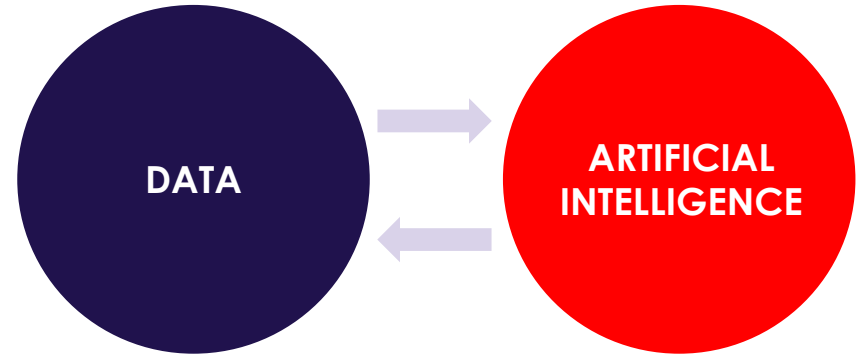
Chat question

Can you think of any big data sources that are relevant to your work?



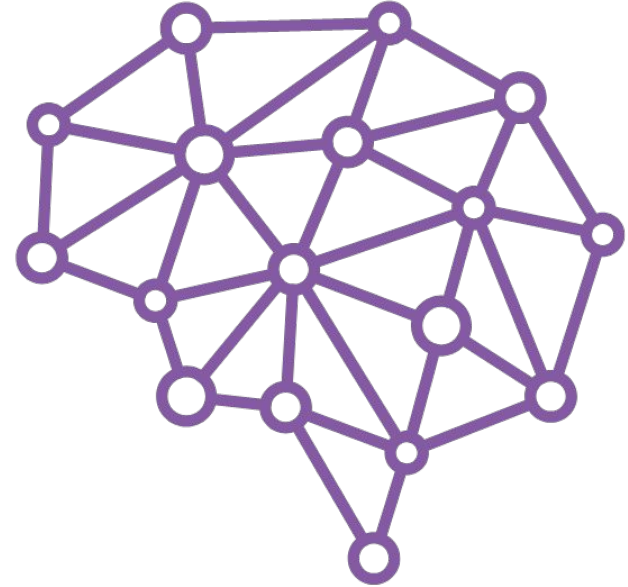
Data and AI

- **Data and artificial intelligence** are **complementary** to each other.
- Data is the foundation of artificial intelligence, providing massive datasets required to train AI technologies.
- AI helps understand and handle data by finding hidden patterns and connections that would be practically impossible for humans to discern from massive datasets manually.



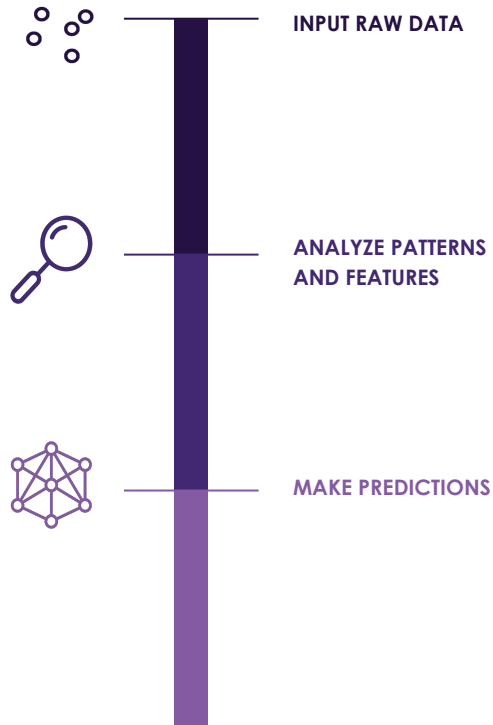
What is AI?

- **Artificial intelligence (AI)** refers to systems or machines that **mimic human intelligence to perform tasks and can iteratively improve themselves** based on the information they collect using machine learning (ML).
- It is a form of intelligence that is used to solve problems, come up with solutions, find answers, make predictions or recommendations.



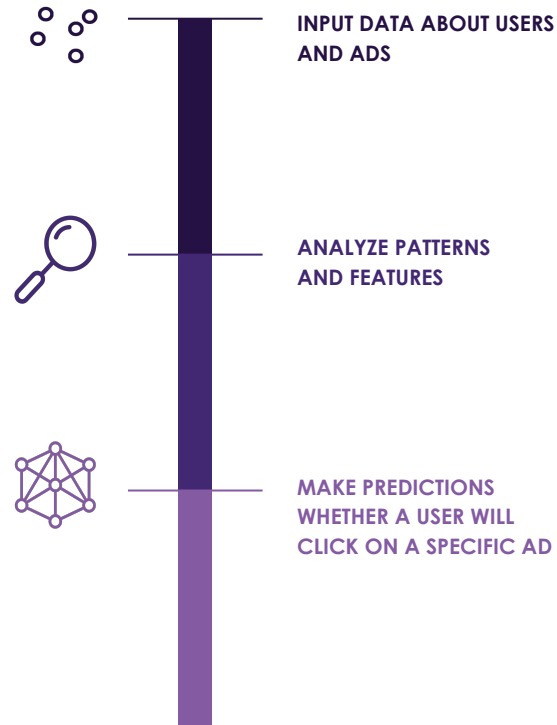
How AI works?

- AI “algorithms” are trained using large datasets so that they can identify patterns, make predictions and recommend actions, much like a human would, just faster and better.
- There are several data science methods, including machine learning, applied to build AI technologies.



Example: How AI works

- Online advertising platforms leverage AI to analyze user data (*demographics, interests, browsing history, etc.*) to understand individual preferences and predict ad engagement.
- This analysis allows platforms to precisely target users with ads tailored to their interests, ensuring greater relevance and effectiveness.



What tasks can AI do?

Handle data
processing
and analysis

Assist in
discoveries
and research

Improve
efficiency
and reduce
costs



Automate
repetitive tasks

Provide
personalized
experiences

AI solution example

- The U.S. State Department has adopted AI technology **to automate** its previously manual process of **declassifying documents** and handling Freedom of Information Act (FOIA) requests.
- To streamline the process, they developed an AI model and trained it on declassification decisions made between 2020 and 2021.
- Using the AI tool, they were able to achieve an accuracy rate exceeding 98% and reduce the manual workload by over 63%.

Sample FOIA request

11/5/2018

Gerard Bardoli
FTA
345 15th Street N
Washington D.C
209-890-0909

Dear Public Official or Agency:

Pursuant to the Missouri Sunshine Law (RS Mo Chapter 610), I would like to obtain a copy of the following public records:

- FTA assignments
- TEI official copy

I understand that if I seek a copy of this record, there may be a copying fee. Please inform me of that cost prior to making the copy. I can be reached at 340-453-0919.

According to the statute, this request should be acted upon as soon as possible, but in no event later than the third business day following receipt of this letter. If access to the records I requested is going to take longer, please contact me so we can work out a reasonable date.

If you choose to deny the request, then you are required to respond in writing and state the statutory exception authorizing the withholding of all or part of the public record and the name and title or position of the person responsible for the denial.

Thank you for your assistance on this matter.

Respectfully,

Tenzin Robert
Phoenix, Arizona

Chat question

Where do you like to grocery shop? And do you like the way they have it set up?



Using data and AI example

- Ever noticed how easy it is to find what you need in a massive supermarket?
- That's no accident!
- These stores use data and analytics to design their layouts to offer a smoother shopping experience to their customers.



Watch the video: [How Walmart is using data](#)

Chat question

Could you share an example of an AI-powered tool you have used, and how you used it?



Break



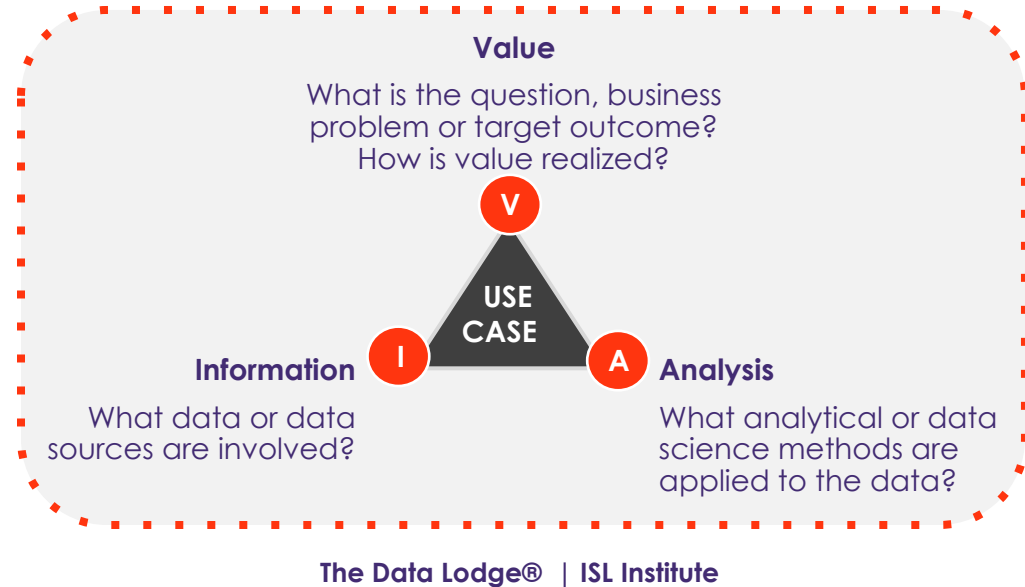
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- What is data literacy?
- Fundamentals of data
- Introduction to big data and AI
- **Data analytics**

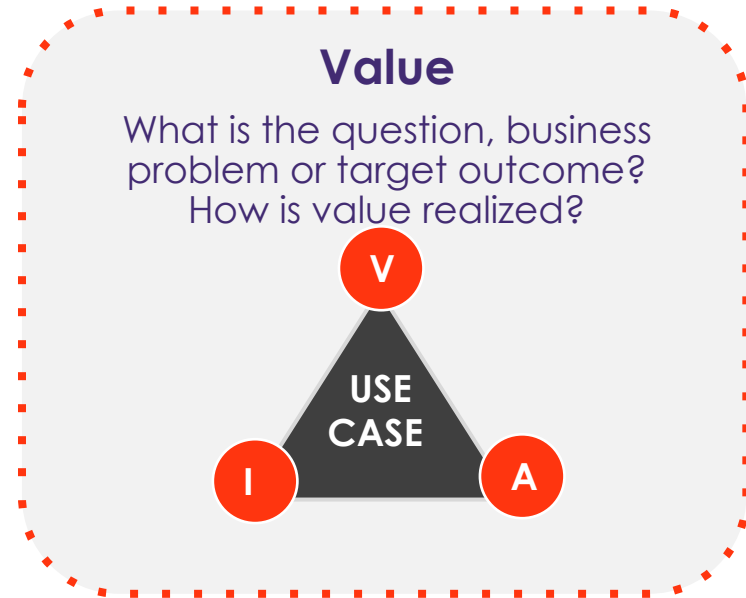
Framework for a data-driven approach

- Data requires a data-driven framework for effective application because data alone is not enough.
- For example, when baking a cake, would you just throw flour, sugar, and eggs into an oven?
- Of course not! You need a recipe, the right ingredients, and the right tools to bake a delicious cake.



1 Value: Defining the business problem

- Defining the problem is **articulating the business objective** and **understanding the key requirements** and constraints, such as available data, budget limitations, and project deadlines.
- In data-driven projects, knowing the why (*the project's purpose*) and its link to business goals is the essential first step.
- A well-defined problem sets the stage for a successful data project that delivers value to the organization.



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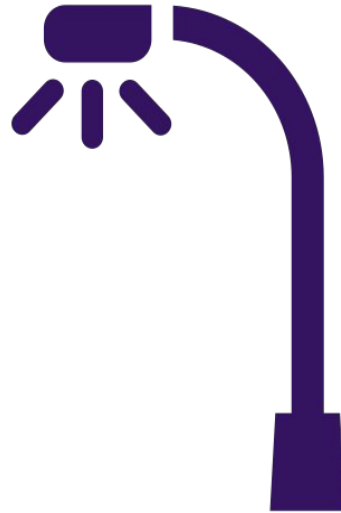
The streetlight effect

The streetlight effect is a term used to describe a situation where people look for information where is it most convenient.



The streetlight effect (cont'd)

- To gain a more comprehensive understanding, we should ask ourselves:
 1. What are you searching for?
 2. Where are you searching?
 3. What is your measure of success?
 4. In what ways could your own assumptions influence your data-gathering process?



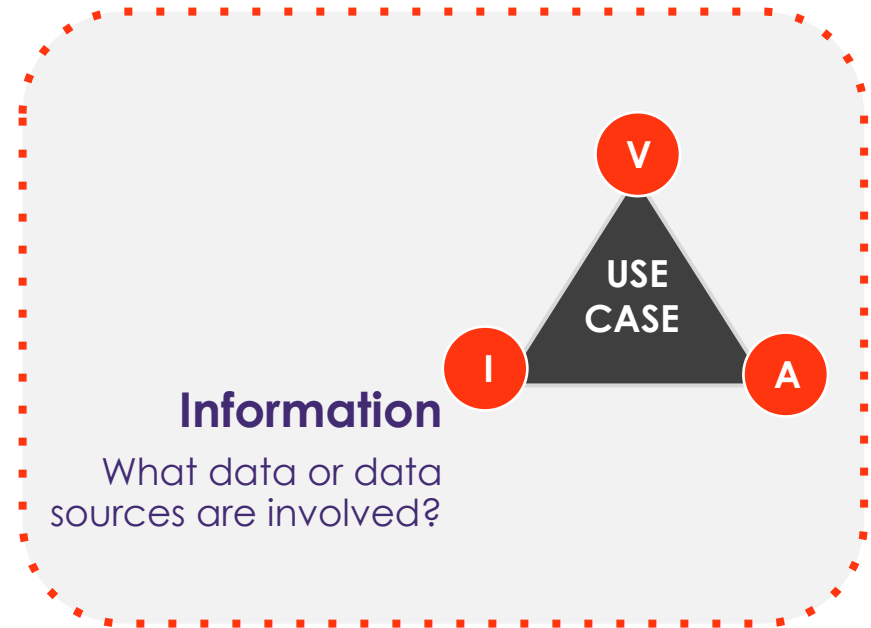
Chat question

How can you look beyond the obvious symptoms to identify the underlying causes of a problem?



2 Information: Finding the right data

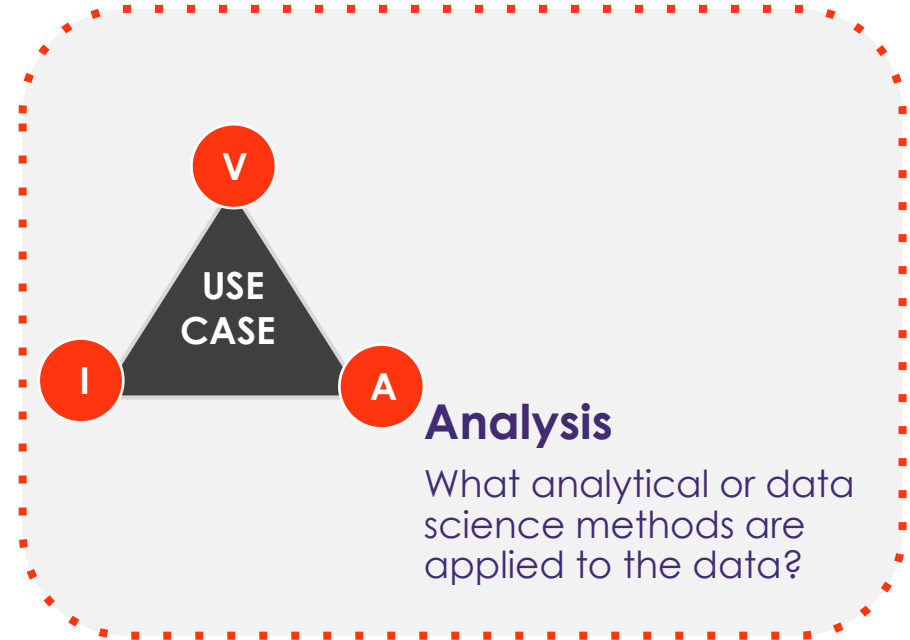
- The next step is to **identify the data** that can help you answer your questions or solve your business problem.
- The data required can be primary data (*collected by yourself or your team or your organization*) or secondary data (*obtained from existing external sources.*)
- It is essential to **consider the reliability, validity, and relevance** of each data source before using it.



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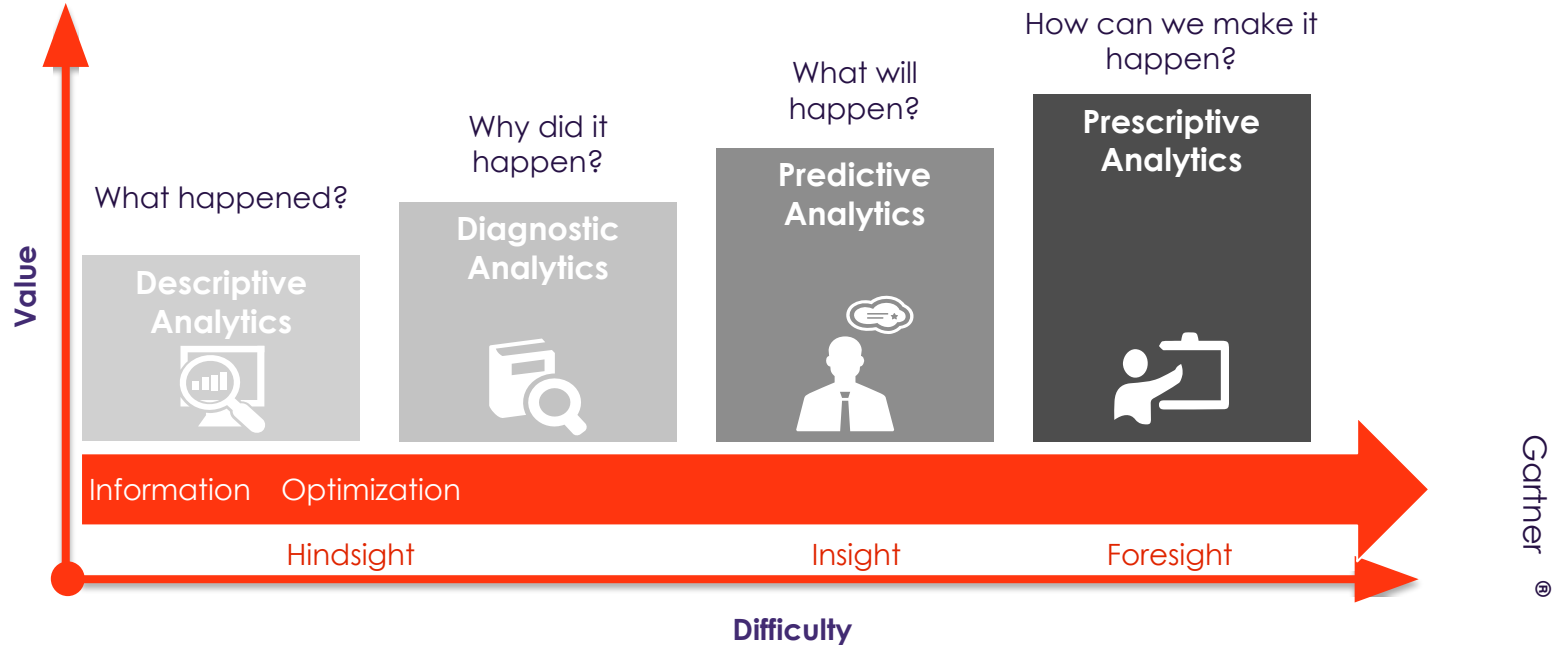
3 Analysis

- **Data analysis** focuses on processing and performing statistical analysis on existing datasets.
- Analysts **capture, process, and organize data to uncover actionable insights** for current problems and establish the best way to present this data.



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Data analytics maturity model



Model revisited

What happened?

**Descriptive
Analytics**



Why did it
happen?

**Diagnostic
Analytics**



What will
happen?

**Predictive
Analytics**



How can we make it
happen?

**Prescriptive
Analytics**



Stage 1: Descriptive analytics



What questions does it answer?	What has happened in the past?
How valuable is it?	Provides some value, but doesn't provide causation or prediction
How labor intensive is it?	Easy to deploy provided you have the right data

Stage 2: Diagnostic analytics



What questions does it answer?	Why did something happen in the past?
How valuable is it?	Provides insights into a particular problem, and can help you identify some root causes for past trends and behaviors
How labor intensive is it?	Requires detailed data, but doesn't have to be overly intensive

Stage 3: Predictive analytics



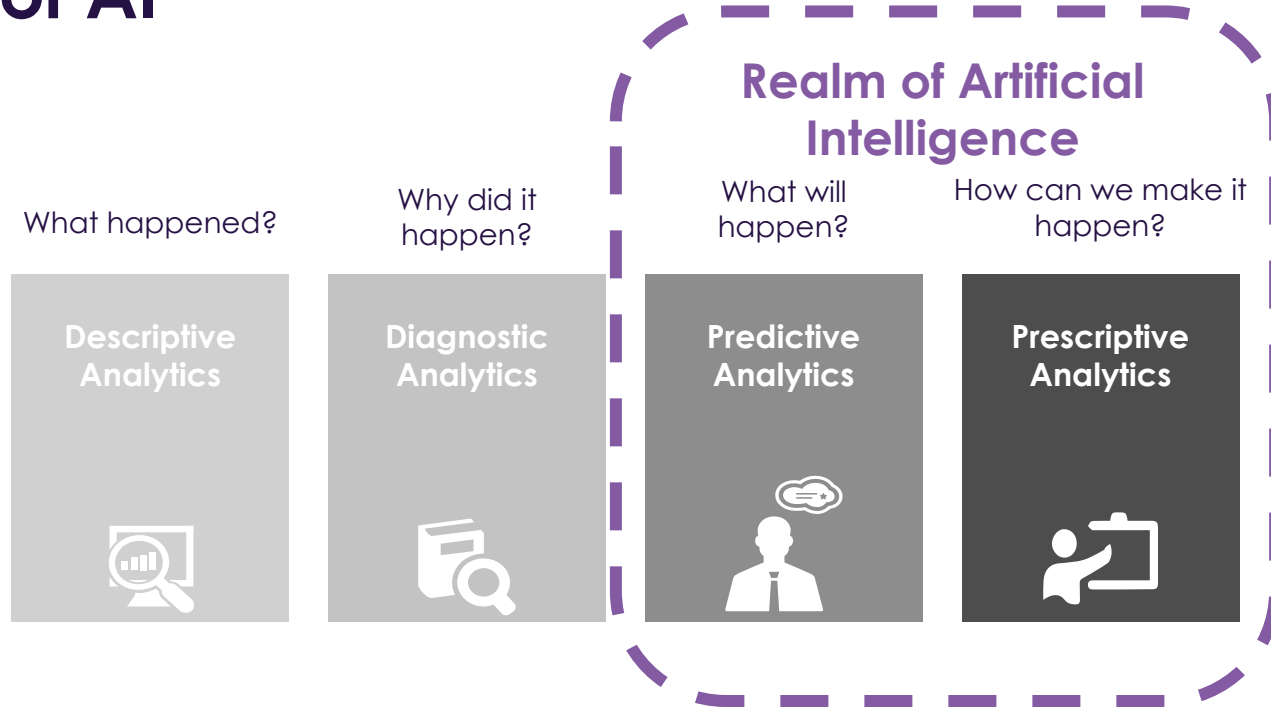
What questions does it answer?	What is likely to happen?
How valuable is it?	Provides trends / behaviors that are likely to happen
How labor intensive is it?	Requires detailed data, and may require a moderate to high level of computer power, depending on the method and the amount of data

Stage 4: Prescriptive analytics



What questions does it answer?	What action should I take next?
How valuable is it?	Provides recommendations for future actions
How labor intensive is it?	Requires a lot of detailed data, as well as data from other external sources that will impact the model; very labor intensive

Realm of AI



Example: FHA

- The Federal Housing Authority (FHA) forecasts default, repayment, and claim rates using big data analytics.
- They also leverage analytics to anticipate the premiums needed to maintain positive cash flow.
- During the 2008 recession, they were the only subprime mortgage insurance fund that did not require bail out.

Example source: [Five Examples of How Federal Agencies Use Big Data \(linked\)](#)



Image source: [Designed by senivpetro on freepik.com](#)



Polling question

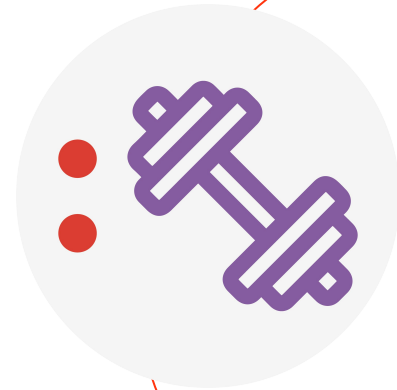
[Select ALL that apply]

What type of data analytics is demonstrated by the Federal Housing Authority example?

- a. Descriptive
- b. Diagnostic
- c. Predictive
- d. Prescriptive

Exercise: Data analytics case study

- Take 10 minutes to analyze the scenario and assign the provided questions to a specific stage in the data analysis journey.
 1. **Descriptive Analytics**
 2. **Diagnostic Analytics**
 3. **Predictive Analytics**
 4. **Prescriptive Analytics**



Refer to **page 6** in the participant guide for this activity.

Debrief time

Share your findings and responses with the whole group.



Key takeaways

- Data literacy and AI readiness are intrinsically linked, like two sides of the same coin.
- Being data literate will equip employees with the skills and knowledge needed to navigate the data-centric world of AI.



Summary

Today, we learned about:

- What is data literacy?
- Fundamentals of data
- Introduction to AI
- Data analytics

Next session, we will learn about:

- Data management
- Data governance
- Data project lifecycle



• End of day 1



DATASOCIETY:

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Day 2



Agenda

Day 2

- **Data management**
- Data governance
- Data project lifecycle

Chat question

What do we mean when we say
data management?



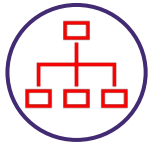
Data management

- **Data management** is defined as the practice of **collecting, processing** and **using data securely and efficiently** for better business outcomes.
- Organizations tailor data management approaches to their specific data needs.
- Small businesses may use simple processes, while large organizations with more complex data needs require a wider range of data management techniques.



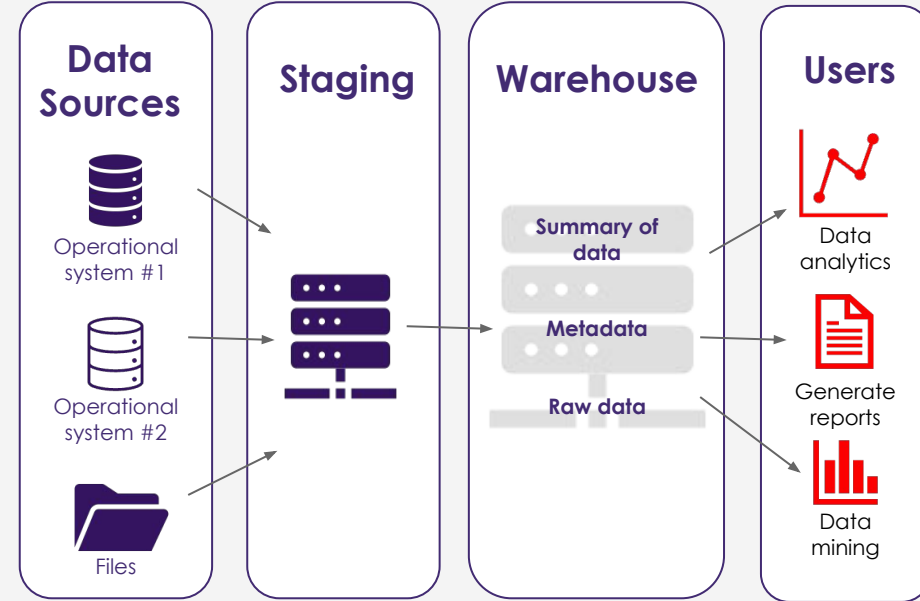
Data management subject areas





Data architecture

- **Data architecture** is the plan for how an organization manages its data.
- It **ensures data is organized, accessible, and secure**, enabling a flexible and scalable data strategy.
- Ideally, it is the first step in data management, but it can be complex, especially in organizations with evolving IT systems.





Data cataloging

- **Data catalog** is a detailed inventory of all data assets in an organization, designed to help employees quickly find the most appropriate data when needed.
- It encompasses important **metadata**, i.e., information that describes different aspects of a dataset in order to:
 - Classify
 - Organize
 - Label
 - Summarize the data

Bank name	ForALL
Month	July
Year	2015
Last updated	2-Dec-15

Metadata

Account	Date	Description	Category	Income Money	Expense Money	Overall Balance
Checking	9/01/17	Beginning B	[Balance]	750		750
Cash	9/01/17	Beginning B	[Balance]	50		800
Credit	9/01/17	Beginning B	[Balance]		340	460
Savings	9/01/17	Beginning B	[Balance]	850		1310
Cash	9/15/17	ATM	[Transfer]		50	1360
Checking	9/15/17	ATM	[Transfer]			1310
Credit	9/20/17	Jay's Auto M	Fuel		34.58	1275.42
Checking	9/22/17	Smith's	Groceries		25.12	1250.3
Checking	9/30/17	Credit Card	Interest Expense		6.5	1243.8
Checking	9/30/17	Credit Card	[Transfer]		374.58	869.22
Credit	9/30/17	Credit Card	[Transfer]	374.58		1243.8

Types of metadata

Descriptive metadata	Structural metadata	Administrative metadata
• Provides information about the content and context of the data • Includes details about the data such as keywords, author, etc. that is unique to the piece of data	• Provides information that helps establish relationship between objects • Includes table of content, section, column names	• Provides information useful in managing resources • Contains information related to governance, access, and security



Polling question

Title:
Keyword:
Creator:

Location:
Date:
Wind turbine:
Instrument:

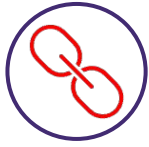
Usage rights:
Project ID:
Project period:

Wake plot of wind turbine
wake plot, multi-rotor, wind
Nikola Vasiljević

Risø, Denmark
14 August 2018
Vestas Multi-rotor
Long-range windScanner

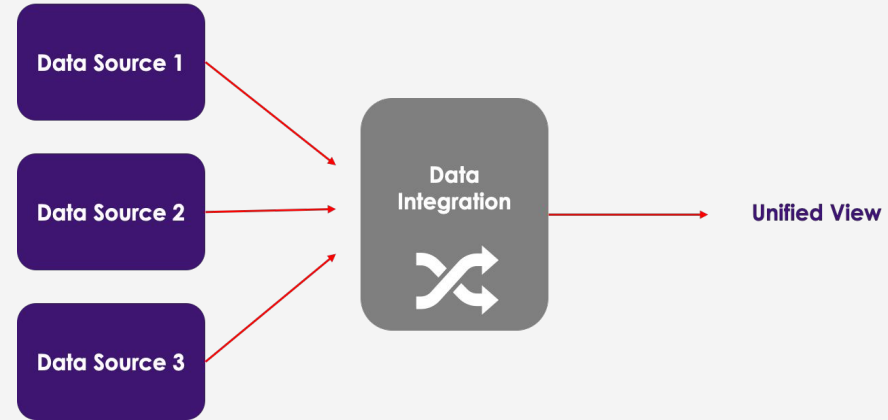
Creative Commons
Vestas_DTU_multi_2018
01 November 2017 – 31 July 2019

1. What type of metadata does the **purple** border outline?
2. What type of metadata does the **orange** border outline?
3. What type of metadata does the **green** border outline?



Data integration

- **Data integration** merges data from multiple sources into a single, **unified view**, enabling comprehensive analysis.
- ETL (Extract, Transform, Load) is a common data integration process designed for preparing data:
 - **Extract data** from different sources
 - **Transform** into a suitable format
 - **Load** into the storage environment



Chat question

- The principle of '**garbage in, garbage out**' is frequently discussed in the context of data and AI. Are you familiar with this concept?
- How do you interpret this phrase?





Data quality

- **Data quality** is crucial as it assesses whether information can serve its purpose in a particular context.

Clean or quality data is:

- ✓ Valid
- ✓ Accurate
- ✓ Consistent
- ✓ Complete
- ✓ Uniform

Clean or quality data is **not**:

- ✗ Corrupt
- ✗ Incorrect
- ✗ Duplicate
- ✗ Incomplete
- ✗ Wrongly formatted

Why insist on quality data?

- Poor-quality data is often pegged as the source of operational snafus, inaccurate analytics, and ill-conceived business strategies.
- Clean data has a range of benefits, and it helps with:
 - ✓ Staying organized
 - ✓ Avoiding mistakes
 - ✓ Improving productivity
 - ✓ Avoiding unnecessary costs
 - ✓ Presenting the true story of the data

Dirty data vs. Clean data example

Incomplete dataset sample

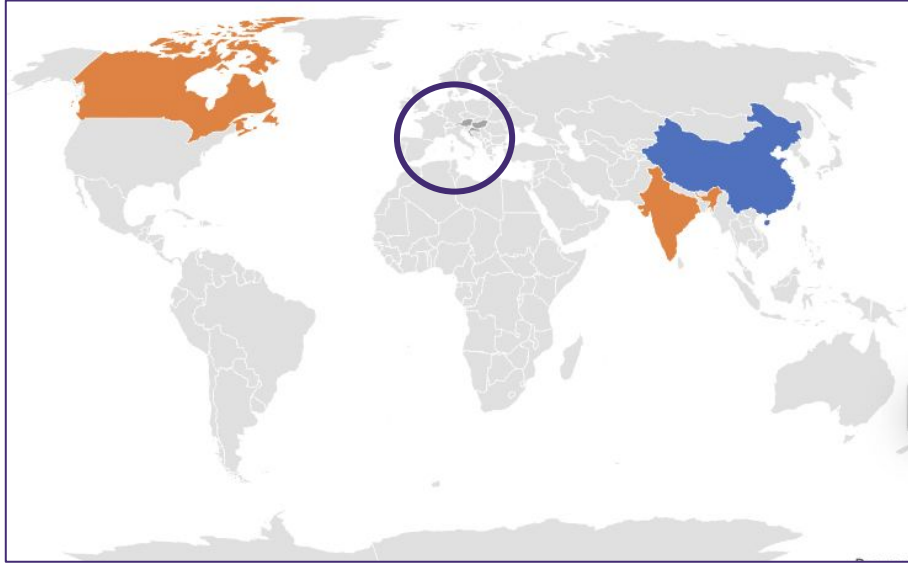
Country	Number of people
China	6
Canada	2
Austria	1
France	
India	2
Hungary	1
Croatia	1

Complete dataset sample

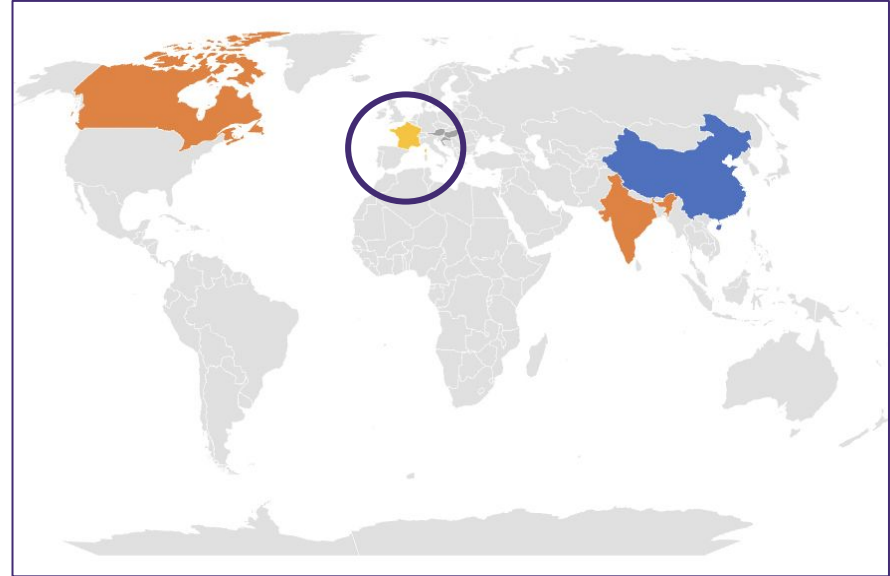
Country	Number of people
China	6
Canada	2
Austria	1
France	4
India	2
Hungary	1
Croatia	1

Dirty data vs. Clean data visualization

Visualizing incomplete dataset



Visualizing clean dataset



Exercise: Detect errors

- Turn to **page 8** of the participant guide to find this activity.
- Take 5 minutes to evaluate the dataset and detect the gaps or inaccuracies in this data.
- Share your answers in the chat.

	A	B	C	D	E	F
1	Name	Last Name	DOB	Country	Country	Status
2	Andrew	Jackson	11/4/87	Argentina	Buenos Aires	Citizen
3	Natalie	Walker	2/12/82	France		
4	Sam	Williams	3/30/80	Argentina	Mendoza	Permanent resident
5	Lilian	Perez	2-Jan-99	France	Nantes	Permanent resident
6	Sarah	Sanchez	1/8/78	Germany	Munich	Citizen
7	J	McAdam	7/18/00	England	London	Permanent resident
8	Oliver	Oliver	12/21/90	Ireland		Citizen
9	Natalie	Walker	2/12/82	France	Paris	
10	Iris	Nelson	10/20/95	India	Mumbai	Citizen
11	Natalie	Roberts	9/30/90	Germany	Munci	Citizen
12	Nina	Rose Turner	1/19/01	UK	London	Citizen
13	Tara	Morris	2/20/63	Delhi	India	Citizen
14	Paul		8/24/03	America	Buenos Aires	Permanent resident
15	PAULINE	BROOKS	7/7/98	India	Banglore	Permanent resident
16	Natalie	Walker	2/12/82	France	Paris	Citizen

Break

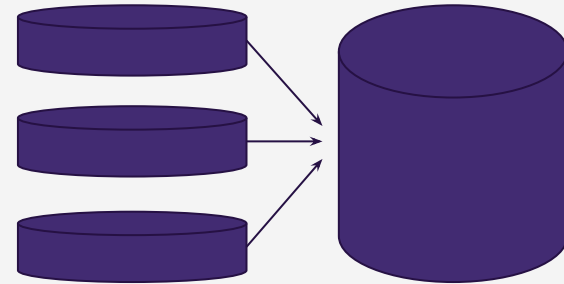




Data storage

- Managing **data storage** helps organizations understand where their data lives.
- There are different ways to store big data:
 - Databases
 - Data warehouses
 - Data lakes

y1	x1	x2	x3
A	F	X	P
B	G	Y	Q
C	H	Z	W



Cloud computing

- **Cloud computing** offers various computing services such as storage, servers, networking, etc., over the internet!
- Common multi-cloud environments are:
 - AWS
 - Google Cloud
 - Microsoft SharePoint
- A multi-cloud environment provides flexibility and matches to diverse requirements.



Traditional

Vs.



Cloud

Chat question

What **storage tools are in use at** your organization?





Data security & access

- **Data access** allows users to get data from any location, in an authenticated manner approved by the organization.
- Organizations are required to set regulations to secure their data and clarify who has access to what data and how.
 - Secure data access prevents data breaches
 - Secure data access reinforce the long term value that data provides an organization



Chat question

What types of sensitive information does your organization handle in its daily operations?



Data security

Data security ensures that:

- Data **privacy and confidentiality** are maintained when dealing with personal, financial, or proprietary data.
- Data is **not breached**, preventing financial losses, business disruptions, and damage to brand reputation.
- Data is **accessed appropriately**, decreasing the potential for legal penalties, reputational damage, and loss of customer trust.



Data security example

- Some employees at Samsung accidentally entered sensitive internal information into a publicly available tool, ChatGPT.
- After discovering this data leak, Samsung banned its employees from using ChatGPT.



Source: [Bloomberg](#)

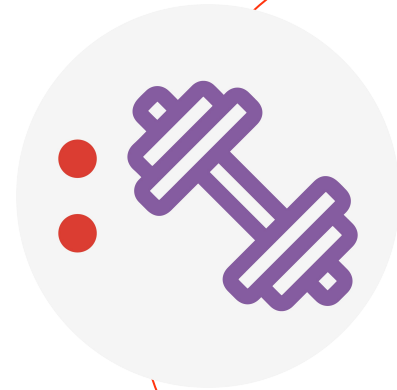
Chat question

How does your organization manage data access across different teams and departments?



Exercise: Data management challenges

- Take 10 minutes to dive into a set of scenarios that highlight data management challenges.
- Analyze each situation carefully and consider how you would approach these issues:
 1. Identify the core data management issues involved in the scenario.
 2. Extract specific pieces of evidence that support the identified issue(s).
 3. Analyze potential consequences of the situation.



Refer to **page 9** in the participant guide for this activity.

Debrief time

Share your findings and responses with the whole group.



Agenda

Day 2

- Data management
- **Data governance**
- Data project lifecycle

Data governance

- **Data governance** is a **set of policies, processes and standards** that help ensure the formal data handling within the organization.
- Within each process, data governance is concerned with:
 - Awareness and communication
 - Policies, standards and procedures
 - Tools and automation
 - Skills and expertise
 - Responsibility and accountability
 - Goal setting and measurement



Key drivers for data governance

1. **Regulatory compliance**

With increased regulation comes compliance that needs to be implemented and followed

2. **Reduce risk**

Effective data governance enhances data security and privacy

3. **Improve processes**

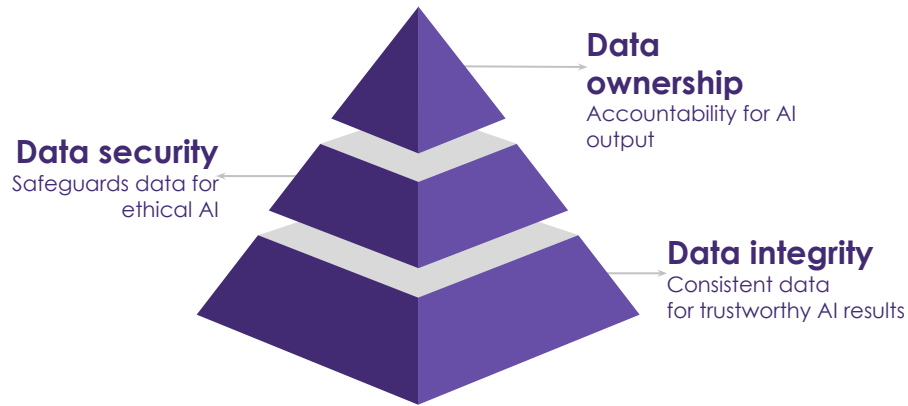
When everyone follows the same standards, projects and management become more efficient

4. **Rise of AI**

There are potential risks related to data privacy, accuracy, and ethical usage

Data governance supports AI

- AI implementation means administration or creation of intelligent agents that can reason, learn, and act autonomously.
- Since AI models rely on data, **data governance plays a significant role in governance in the AI era.**
- It's a strategic advantage in the AI era as it ensures:
 - Quality assurance
 - Ethical considerations and regulatory compliance
 - Scalability



Examples of existing regulations

- National and international governments draft, publish, and enforce data ethics rules.
- Some of the most widely recognized data governance guidelines in the world include:
 - California Consumer Privacy Act (CCPA)
 - Health Insurance Portability and Accountability Act (HIPAA)
 - Family Educational Rights and Privacy Act (FERPA)
 - EU Data Governance Act (DGA)
 - General Data Protection Regulation (GDPR)



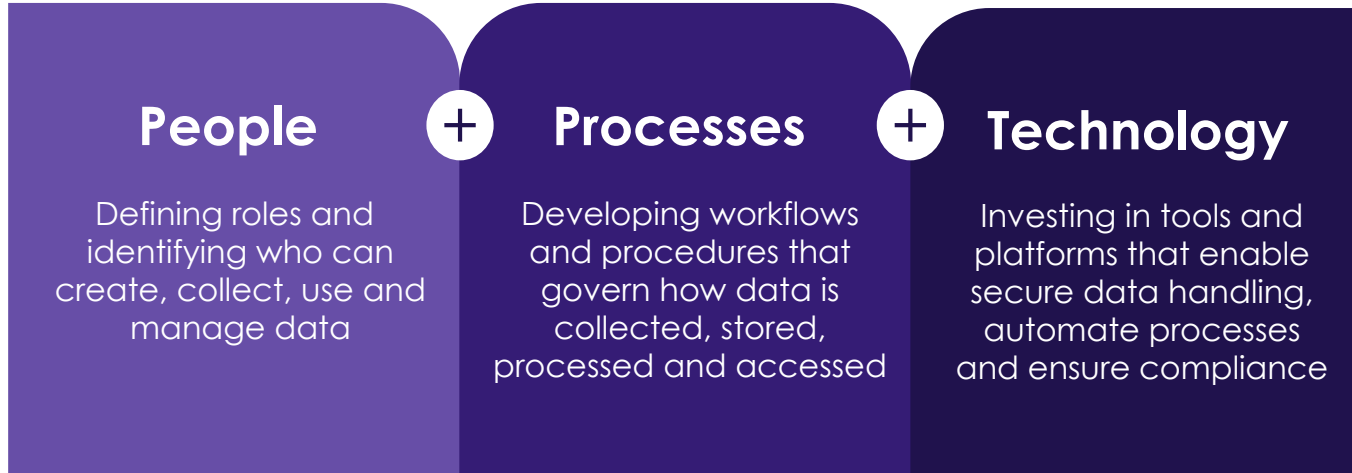
FERPA

Family Educational
Rights & Privacy Act

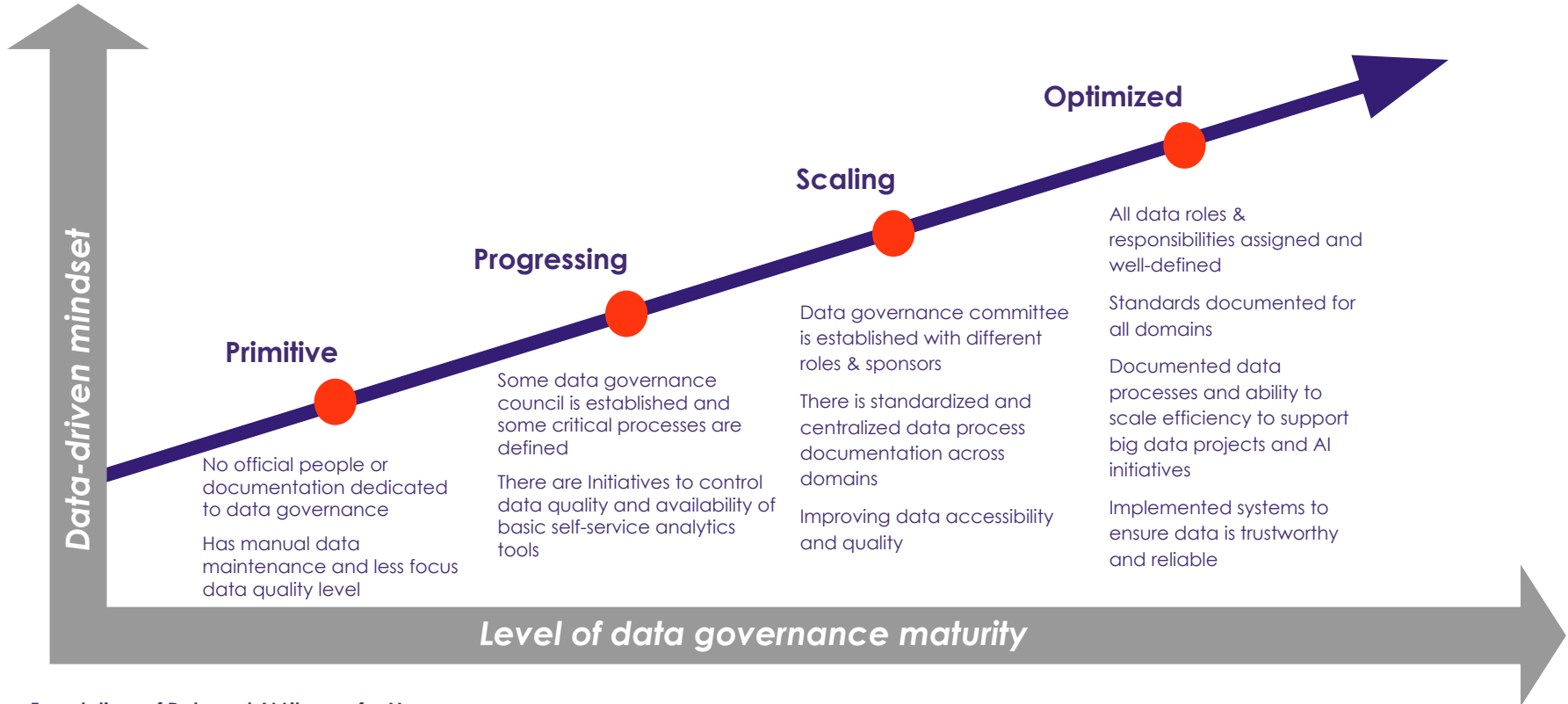


Implementing data governance

- Data governance relies on a balanced approach, encompassing not just policies and standards on processes, but also the people and technology.

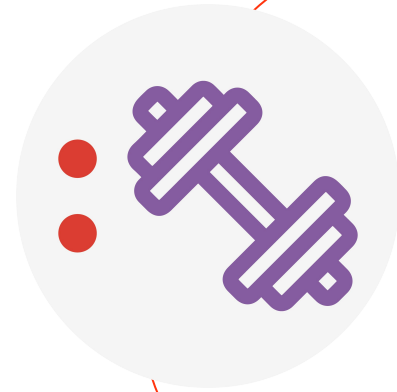


Data governance maturity levels



Exercise: Data governance maturity assessment

- Use the Data Governance Maturity Assessment in the participant guide to evaluate your organization's current capabilities.
- Reflect on your own experiences and observations to provide honest and accurate answers.
- This will help you pinpoint your organization's strengths and areas for improvement in data governance.



Refer to **pages 15-17** in the participant guide for this activity.

Debrief time

Share your findings and responses with the whole group.



How can managers contribute?

- ✓ **Catalog the data** “owned” by your team and employ version control, backups, and redundancy
- ✓ Implement preventative and detective controls to **improve data quality**.
- ✓ **Define roles and responsibilities:**
 - Data owners are accountable for the state of the data.
 - Data stewards make sure that data policies and standards are adhered to and stay abreast of changes.
- ✓ Develop **standardized data definitions** and educate stakeholders on them.
- ✓ **Security!** Track who has access to the data and who has the permission to modify it.
- ✓ Educate your team on the existing data governance policies and standards.

Break



Agenda

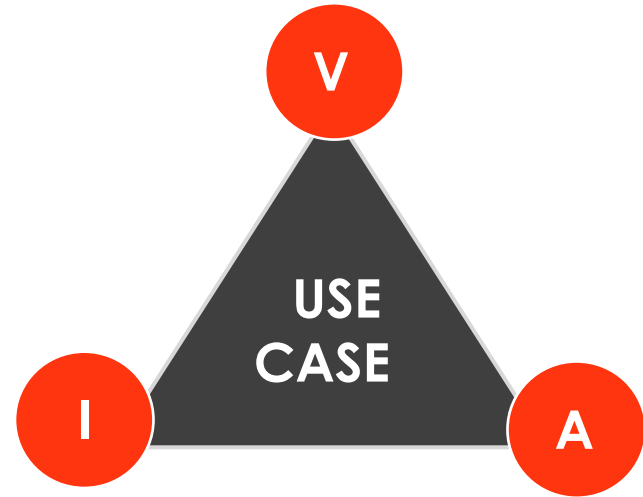
Day 2

- Data management
- Data governance
- **Data project lifecycle**



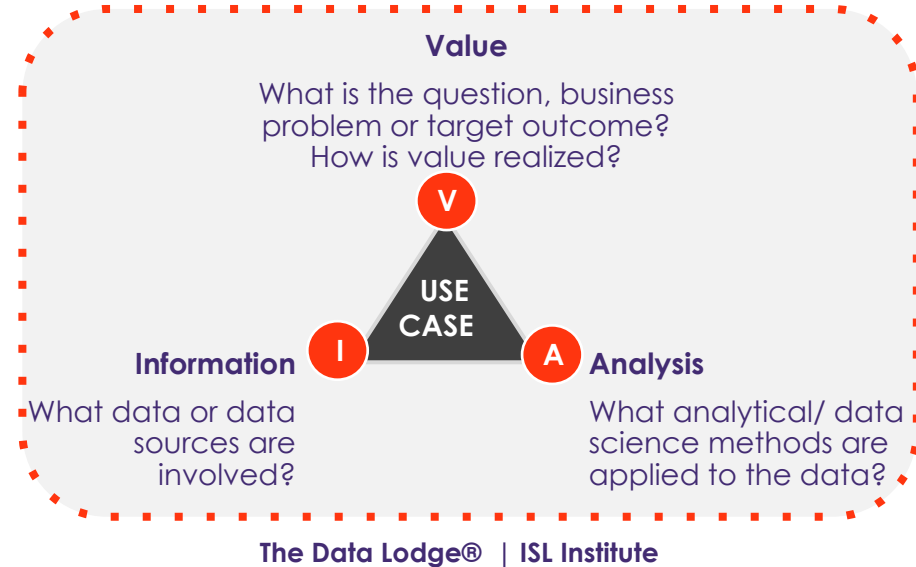
Polling question

Can you recall what the V, I, and A stand for in the VIA model we discussed on Day 1?



Data project

- A **data project** focuses on extracting knowledge and insights from data.
- Statistical methods, data visualization, and programming are used to understand past trends, identify patterns, and make informed decisions.
- When planning data projects, consider the **Value-Information-Analysis** framework which helps connect the dots between a **business issue and its impact**, guiding through the necessary information and analysis.



Typical data project lifecycle

1 What is the problem(s) we need to solve?
Ask

2 What data is needed & how to collect and prepare data?
Research

6 How can we use the conclusions in the real world?
Interpret

3 Which method(s) is appropriate to use?
Model

5 How does the model generalize to real-world data?
Test

4 Do the model and assumptions work as expected?
Validate

Review: Ask and Research

1

What is the
problem(s) we
need to solve?

Ask

2

What data is needed &
how to collect and
prepare data?

Research

6

How can we use
the conclusions in
the real world?

Interpret

3

Which method(s)
is appropriate
to use?

Model

5

How does the
model generalize
to real-world data?

Test

4

Do the model and
assumptions work
as expected?

Validate

Setting KPIs

- **Key Performance Indicators** (KPIs) are **quantifiable measurements** that reflect the critical success factors of a data project and guide future actions.
- Once you understand the "what" and the "why" of the project, you can choose KPIs to effectively measure progress towards those goals.
- Use the SMART framework to make sure your KPIs are strong and drive the results.

Specific

Provide clear, detailed description of what you want to achieve

Measurable

Include a quantifiable metric with a target that evaluates success

Achievable

Set a challenging target, but keep it within the scope

Relevant

Clarify that the KPIs are relevant to your organization and team

Time-bound

Set a deadline for the task/action to be accomplished

Types of KPIs

- When managing data science projects with non-technical teams, the following KPI types emphasize outcomes, impact, and progress:
 - **Operational efficiency KPIs**
Example: Achieve 90% of deliverables on time in the first quarter.
 - **Accuracy KPIs**
Example: Receive a 4.5/5 rating on the ease of using the new forecasting tool.
 - **Performance KPIs**
Example: Reduce report preparation time from 5 hours to 30 minutes per week.
 - **Financial impact KPIs**
Example: Reduce delivery costs by 15% through route optimization.

Chat question

Refine the KPIs by making them specific, measurable, achievable, relevant, and/or time-bound and share them in the chat.

KPI #1:

Reduce employee turnover rate.



Chat question

Refine the KPIs by making them specific, measurable, achievable, relevant, and/or time-bound and share them in the chat.

KPI #2:

Boost brand awareness by 25%.



Chat question

Refine the KPIs by making them specific, measurable, achievable, relevant, and/or time-bound and share them in the chat.

KPI #3:

Encourage data-driven decision-making in the team.



Model, Validate and Test

1 What is the problem(s) we need to solve?
Ask

2 What data is needed & how to collect and prepare data?
Research

6 How can we use the conclusions in the real world?
Interpret

3 Which method(s) is appropriate to use?
Model

5 How does the model generalize to real-world data?
Test

4 Do the model and assumptions work as expected?
Validate

Interpret

1 What is the problem(s) we need to solve?
Ask

2 What data is needed & how to collect and prepare data?
Research

6 How can we use the conclusions in the real world?
Interpret

3 Which method(s) is appropriate to use?
Model

5 How does the model generalize to real-world data?
Test

4 Do the model and assumptions work as expected?
Validate



Polling question

Which phase do you think takes the longest?

1 What is the problem(s) we need to solve?
Ask

2 What data is needed & how to collect and prepare data?
Research

6 How can we use the conclusions in the real world?
Interpret

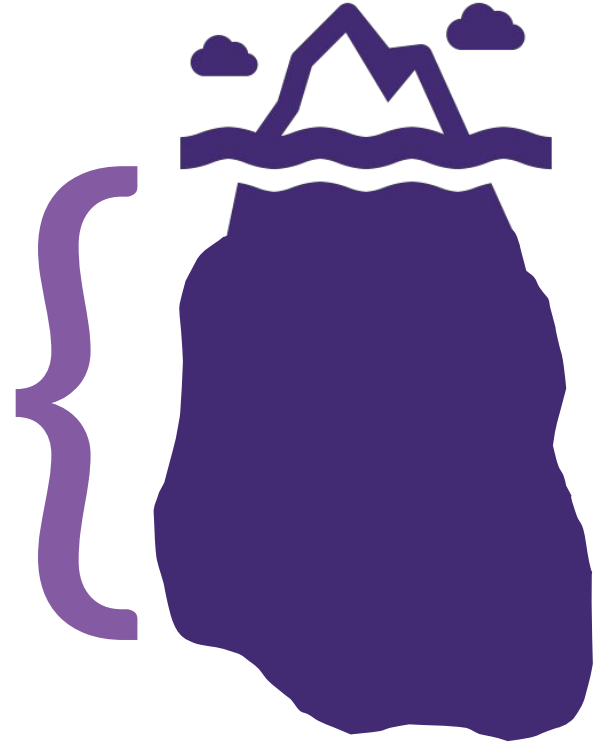
3 Which method(s) is appropriate to use?
Model

5 How does the model generalize to real-world data?
Test

4 Do the model and assumptions work as expected?
Validate

The iceberg of data analysis

The “Research” stage
because gathering and
preparing data
(*cleaning data*) often
takes **70-80%** of project
time.



Discussion question

- What potential project ideas or opportunities do you see where you could leverage data and analytics to track key metrics, measure performance, or improve productivity and outcomes?
- Share your responses in the chat.



Summary

Today, we learned about:

- Data management domains
- Data governance and its significance
- Data project lifecycle

Next session, we will learn about:

- AI for data analytics
- AI implementation



• End of day 2



DATASOCIETY:

Foundations of Data and AI Literacy For Managers

Day 3



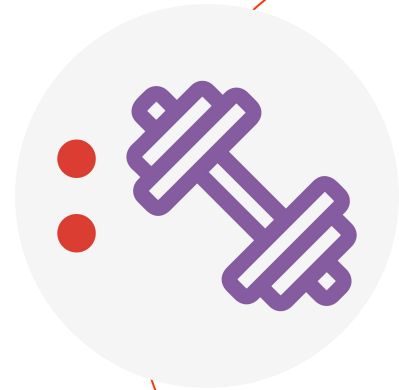
Agenda

Day 3

- **AI for data analytics**
- AI implementation

Exercise: Field trip

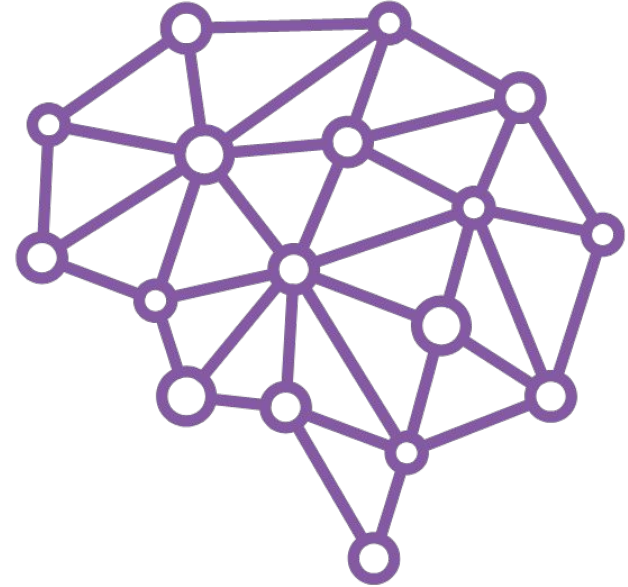
- Go to [Amazon.com \(linked\)](#) and search for a product.
- Take a few minutes to read the “Customers say” summary and key words.
- Chat questions:
 - How does this generated summary impact your experience as a potential Amazon customer?
 - How might you benefit from a similar text summarization tool in your own work?



Refer to **page 16** in the participant guide for this activity.

Review: What is AI?

- **Artificial intelligence (AI)** refers to systems or machines that **mimic human intelligence to perform tasks and can iteratively improve themselves** based on the information they collect using machine learning (ML).
- It is a form of intelligence that is used to solve problems, come up with solutions, find answers, make predictions or recommendations.



AI examples in the finance sector



Enova uses AI in its lending system, **to provide analytics about finances and credit.**



Ocrolus uses AI and machine learning along with human verification **to speed up the analysis of financial documents.**

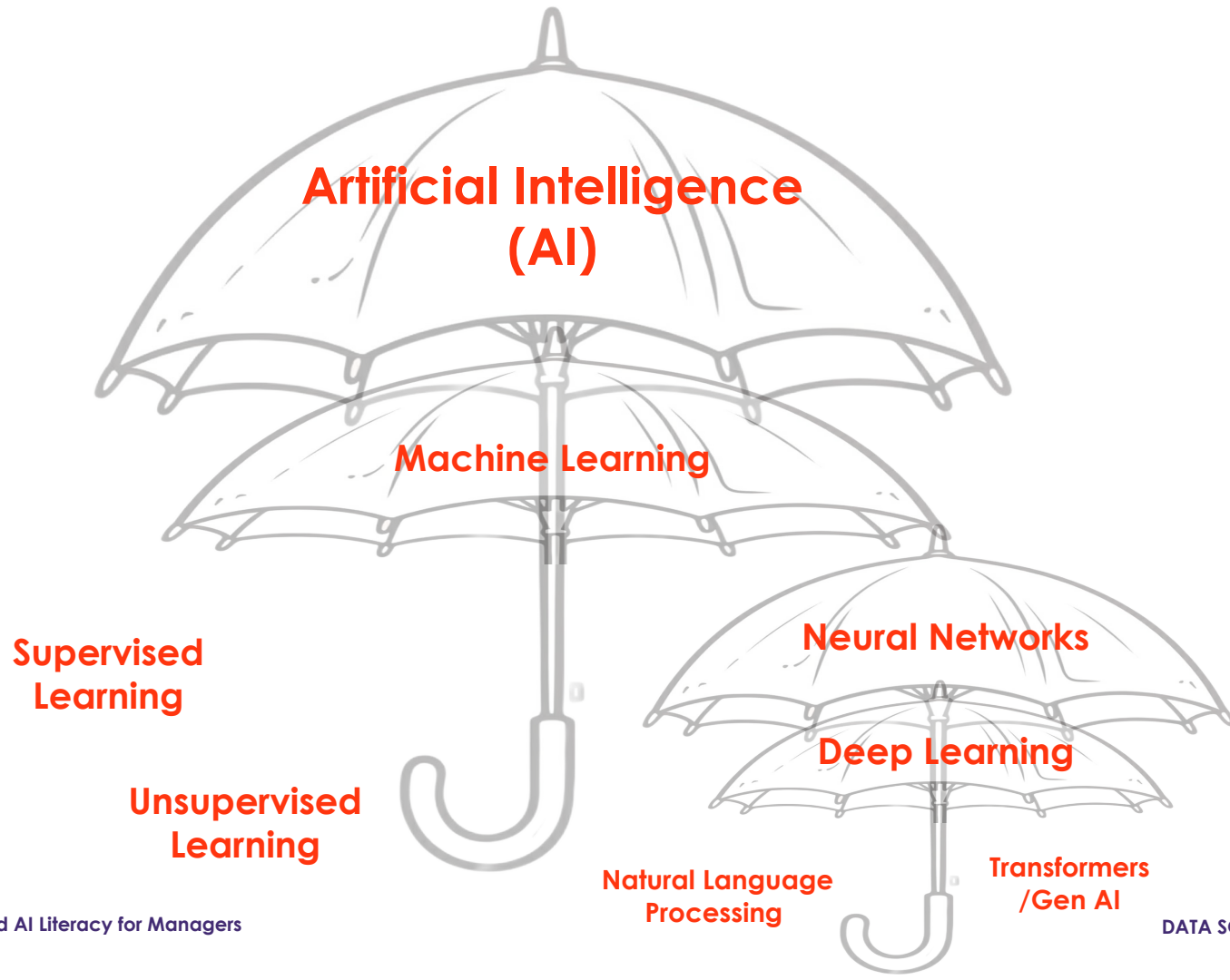


Socure uses AI and machine learning **to verify identity.**

Chat question

- For Enova, Ocrolus, and Socure, what data might be used to fuel their uses of AI?
- How would the outputs here increase efficiency?





Why learn about these concepts?

To cultivate a mindset

Understand what options are available to optimize workflows, and drive success in your projects.

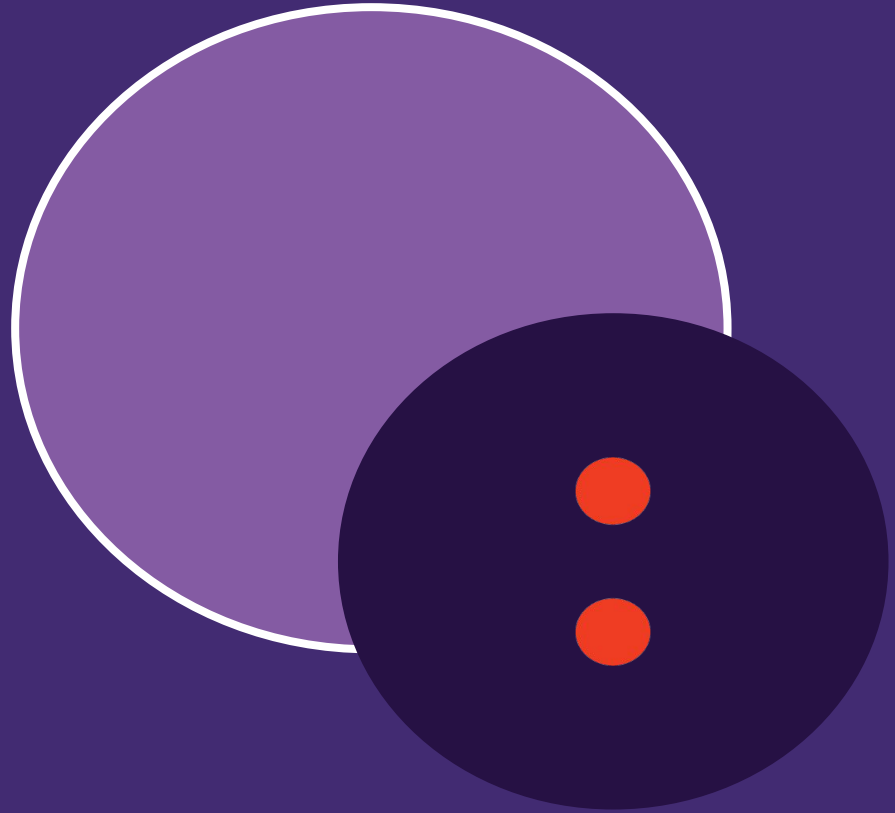
To develop language

Gain basic technical understanding of AI systems and develop vocabulary and fluency in the “shared language”.

To enhance skills

Refine your communication and collaboration skills, especially when working with data science teams.

Machine learning



What is machine learning?

- **Machine learning (ML)** uses **algorithms** to find patterns in massive amounts of data and **predict** future results with minimal human intervention.
- It powers many of the services we use today:
 - Recommendation systems like those on Netflix
 - Search engines like Google
 - Voice assistants like Siri and Alexa
- Most is categorized as either supervised, unsupervised or semi-supervised.



Machine learning: examples

Input 	Algorithm (Set of steps to accomplish a task)	Output 
Emails	Spam filtering functions	Email is a spam or not
Text in English	Machine translation functions	Text in French
Sequence of words	Functions to understand the meaning of each word and how they relate to each other	Recommend words to complete the sentence
Audio data	Speech recognition functions	Text transcripts
Stock market data (price, volume, etc.) over time	Predictive modeling algorithms	Forecast of future stock prices

Supervised learning

- An algorithm is given **tagged and labeled** input data to create such a solid map that we can predict the output for any novel input data.
- The goal is to map known input to a known output.
- For example, in predicting house prices, a human labels data and includes features like time of year, interest rates, square footage, etc. and specifies the output, i.e., the housing prices. The algorithm learns to identify patterns and correlations between the input and output and takes the features of a new house and predict its likely price.



Unsupervised learning

- An algorithm is allowed to detect and learn patterns based on **untagged** input data, usually to **generate categories**.
- The goal is to map known input to a unknown output.
- For example, by analyzing website customer journey data such as pages visited, clicks, etc., the algorithm autonomously identifies patterns and groups users with similar behavior.



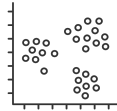
Chat question

Can you think of any AI/ML applications in your industry by other organizations?



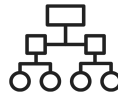
ML foundational methods families

- Machine learning provides **a diverse set of methods for data analytics**, and projects often involve strategically applying multiple methods to address the challenges.



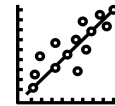
CLUSTERING

generating labels
from unlabeled
data



CLASSIFICATION

applying labels to
novel data points

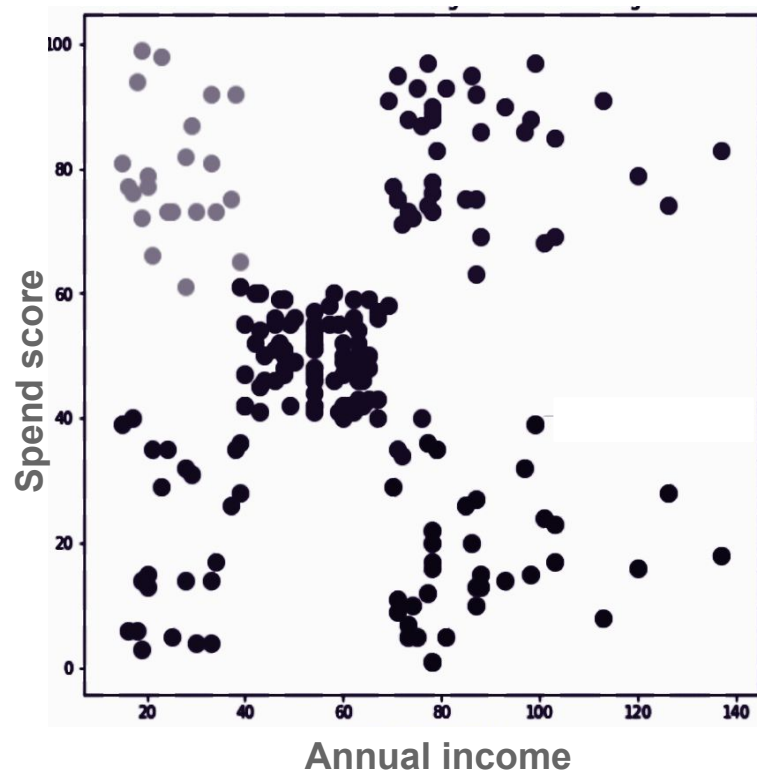


REGRESSION

predicting the
influence of various
factors

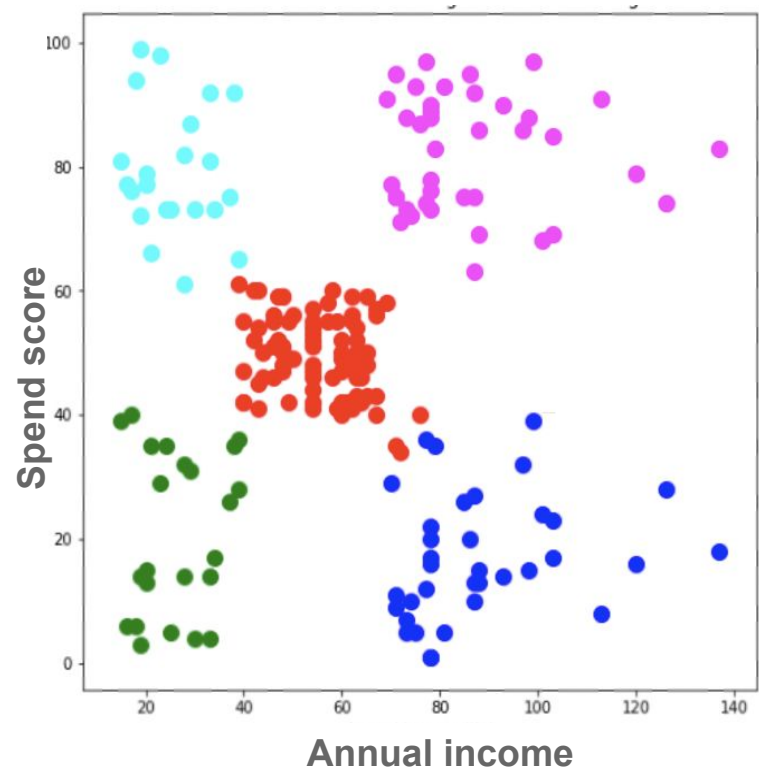
Clustering example

- **Clustering** is a way of grouping data points together based on their similarities and discovering hidden groupings in data.
- Since these groups **aren't labeled or predefined**, it requires domain knowledge to interpret or extract insights.
- For example, a company uses clustering to identify previously unrecognized customer segments within their sales data and target advertising accordingly.
- Chat question: **Do you observe any customer segments emerging?**



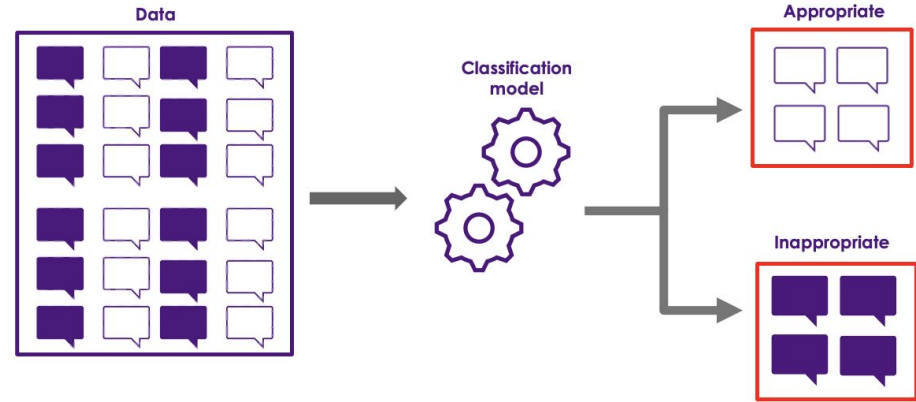
Clustering example (cont'd)

- Collaboration with domain experts helps define clusters, enabling clearer grouping, interpretation, and decision-making.
- The graph presented illustrates a potential visualization of the defined clusters.
- Chat question: **Which cluster would you market a new luxury car to?**



Classification example

- **Classification** is used for identifying and grouping objects or ideas into **predetermined categories**.
- For example, LinkedIn uses a classification model to sort its posts as appropriate or inappropriate.
- The model is trained on a dataset of posts labeled as "appropriate" or "inappropriate" and sorts new posts based on similarity to the previously seen posts.



Regression example

- **Regression** helps **predict the value** of a variable based on the value of another variable(s).
- For example, when browsing homes online, you have likely encountered estimated property values on popular real estate platforms like Redfin and Zillow.
- These platforms leverage ML techniques like regression to predict the fair market value of homes.



ML advanced methods families

- Advanced machine learning methods build upon fundamental concepts to tackle complex problems and achieve higher accuracy and efficiency.



TEXT MINING

inference and
prediction based on
textual data



GRAPH ANALYSIS

finding patterns within
a network to show
connections

Text mining example

- **Text mining** helps analyze **unstructured text data** to extract meaning and identify patterns.
- For example, researchers used text mining to analyze tweets about different smartphone brands, like #iPhone and #Samsung, to understand consumer sentiment and trends.
- This allowed them to extract valuable information from the unstructured text data of Twitter.

Document info

ID	Title	Position	Key word
123	The Catcher in the Rye	6	Testing
334	Pride and Prejudice	4	behind
432	Brave New World	2	all
5854	Alice's Adventures in Wonderland	5	wonder
153	The Adventures of Huckleberry Finn	34	experience
64	To Kill a Mockingbird	7	sky
737	1984	1	willingly
847	Wuthering Heights	10	bio
790	Jane Eyre	12	society
908	The Lord of the Rings	3	real
8664	The Sun Also Rises	9	every

Graph analysis example

- **Graphical analysis** uses visual representations like graphs and charts to **identify patterns, trends, and relationships between variables** within a dataset.
- For example, Google Shopping uses your searches and activity to identify related products you may be interested in.
- This It uses patterns and relationships to make personalized recommendations.

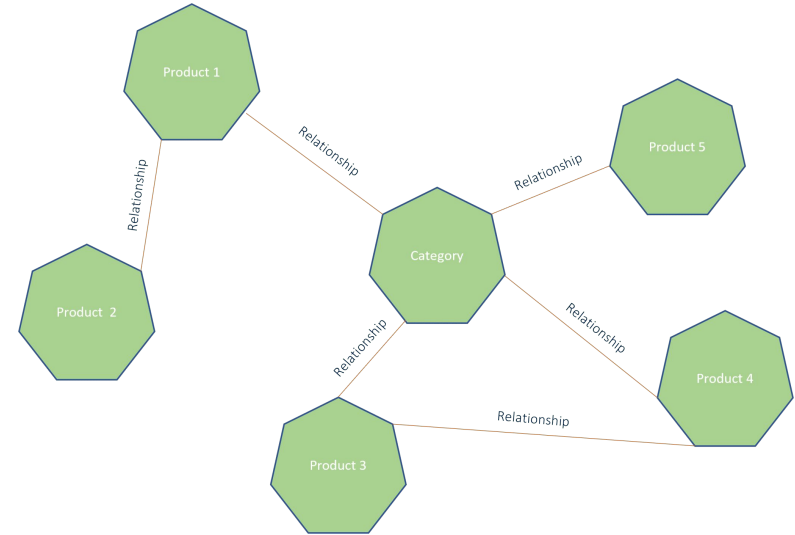


Image source: [Kopp Online Marketing Consultant](#)

Chat question

Which concept(s) resonated with you the most, or do you have some experience in any of the areas discussed?



Break



Agenda

Day 3

- AI for data analytics
- **AI implementation**

AI initiatives

- **AI projects** encompass the **implementation or creation of intelligent agents** that can reason, learn, and act autonomously.
- AI often leverages data science techniques but goes further by building models that can make decisions or perform tasks without explicit human instruction.
- ML is a subfield of AI where algorithms learn from data to improve their performance on a specific task.

AI project examples

AI chatbots



Self-driving cars



Music recommendation apps



Automatic text summarization



AI project lifecycle

- Similar to the data project lifecycle, there are generally 6 stages involved in building AI systems.
- If your organization embarks on an AI project, as a manager, you can contribute by:
 - Facilitating **collaboration** between teams
 - **Identifying and removing** any organizational obstacles
 - Ensuring the project focuses on delivering **measurable value** and achieving defined objectives



AI project requirements

- Building AI requires vast amounts of carefully curated data that is:
 - Relevant real-world data
 - Properly classified/labeled data
 - Formatted data
 - Meets minimum volume requirement
- Additionally, **pre-trained models** and **robust computing power** can also be helpful for efficient and effective model development process.



Image source: Created using generative AI tool

Why does AI need so much data?

- AI models heavily rely on their training data as they **learn patterns** from that training data and apply those patterns to new, unseen data.
- Their performance is directly tied to the quantity and quality of the training data.
- The reasoning capabilities of AI are purely inductive and it may struggle with situations that are significantly different from its training data.

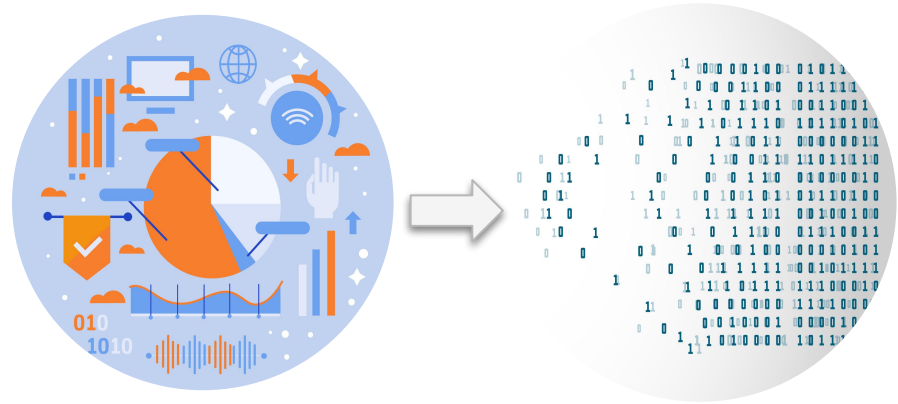
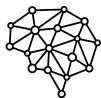


Image source: [Designed by starline at Freepik](#)

AI methods

- There are three fundamental methods that drive **AI development** towards achieving human-like intelligence.



NEURAL NETWORKS

Modelled on the human brain, these algorithms train on large amount of data and have built in mechanisms to reduce modelling errors



DEEP LEARNING

Composed of neural networks that have more than 2 hidden layers, it is commonly used in text and image prediction



TRANSFORMERS

Designed to process sequential NLP data, these neural networks have an added encoder and decoder layers

AI methods for computer vision

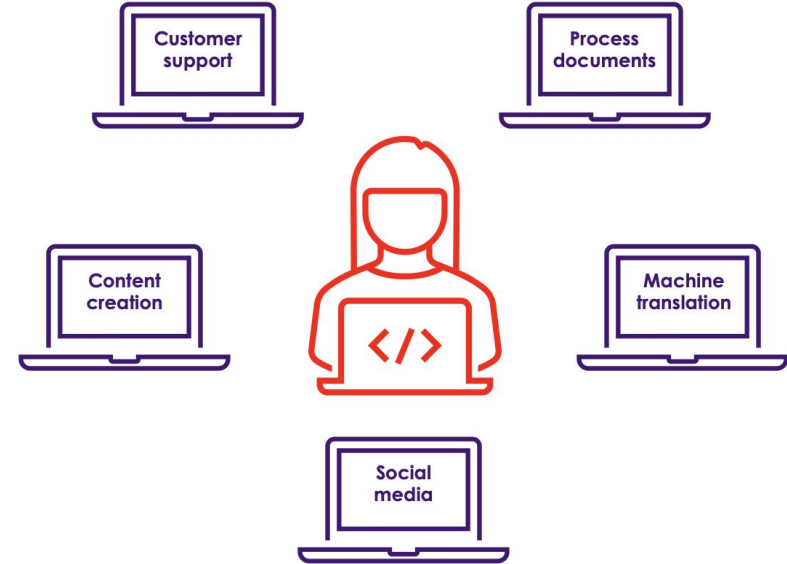
- **Computer vision**, a field of AI, applies neural networks and deep learning to understand and make decisions about images and videos.
- For example, in facial recognition systems, the computer sees only a grid of pixels, each with a specific brightness value.
- Here a neural network's task is to analyze the grid of numbers, decipher the patterns within, and ultimately identify the person in the image.



Source: [Designed by Freepik](#)

AI methods for LLMs

- **Large language models (LLMs)** are a type of deep learning model designed to understand and generate human language.
- They represent **just one category of generative AI** as they focus specifically on text generation.
- For example, virtual assistants like Siri, Alexa, Google Assistant use LLMs to process natural language queries and provide information or execute tasks.



Chat question

What AI tools are you using at work, and how are they impacting your productivity?



AI in data management

- With AI, organizations can build more efficient data management processes that are less prone to errors.
- Many types of **data management tools now incorporate AI and ML capabilities**, for example:

Data collection

- Classify data automatically based on predefined rules or machine learning models
- Extract structured data from unstructured data sources

Data cleaning

- Perform validation checks
- Flag or correct errors, improper formatting and irregular value
- Convert data to the appropriate format
- Detect duplicates

Data storage

- Scan network devices and data storage repositories
- Enables rapid retrieval and processing of data

Data access/security

- Automate creation of data catalogs
- Apply security controls and flag or block unauthorized use of that data
- Analyze patterns and spot abnormal transactions

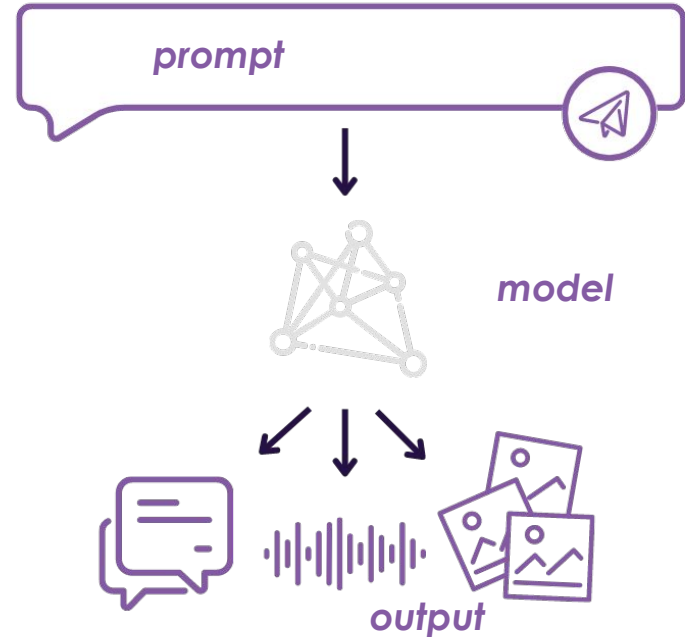
Chat question

- While we have talked about AI and its subfields, there is one more fascinating branch called "Generative AI."
- What have you already heard or learned about Generative AI?



Generative AI

- **Generative AI** is a category of artificial intelligence that **produce new, realistic outputs** in the form of text, images, audio, video, or synthetic data.
- Typically, it starts with a simple prompt in which the user describes the output they want.
- Then, various algorithms generate new content according to what the prompt was asking for.



Types of content generation



TEXT

[ChatGPT](#)
[Google Gemini](#)
[Microsoft Copilot](#)



IMAGE

[Adobe Firefly](#)
[DALL-E-3](#)
[Mid Journey](#)
[Stable Diffusion](#)



AUDIO

[Murf AI](#)
[MuseNet](#)
[Play HT](#)
[Soundraw](#)

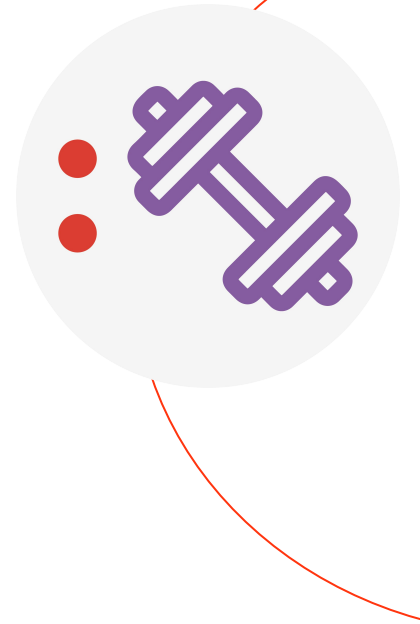


VIDEO

[Lumen5](#)
[Synthesia](#)

Instructor demo: Image generation

- Let's briefly experiment with an image generator.
- Submit **your image idea** that you would like to create in the chat (be creative!).



Build vs. Buy AI

- Organizations dwelling into AI have a pivotal decision to make: **to build AI toolsets in-house or to buy them from external providers?**
- While buying AI offers convenience and affordability, building in-house provides the flexibility to tailor the system to your specific needs and gain a competitive advantage.
- The market for AI tools is exploding which includes:
 - Easy-to-use AI building tools
 - AI models as a service
 - AI plugins for existing software

Chat question

- To build or to buy? Many organizations are facing this question today.
- What influences their decision? What questions are crucial to explore?



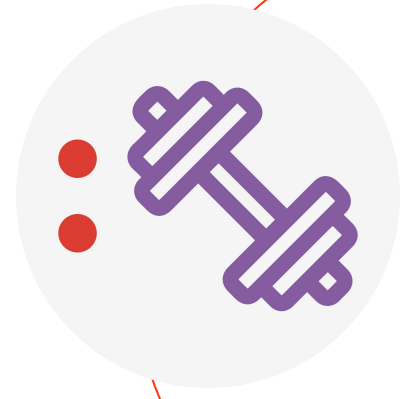
Questions to consider

- ✓ What are the **benefits** of implementing AI?
- ✓ What are the **total costs** of building an AI system ourselves, compared to buying one that already exists?
- ✓ Do we have the **resources** (*data, skills, and time*) to create an AI system that outperforms existing ones?
- ✓ Which approach – building or buying AI – is more likely to keep our company **compliant with upcoming AI regulations**?
- ✓ What are the **data privacy challenges** related to buying vs. building?
- ✓ What **risks** could our organization face if purchase of an AI system leads to a vendor lock-in?

Exercise: AI brainstorm

Take 10 minutes to explore the potential applications of AI in your work domain and jot down thoughts on the following questions:

- **What are some of the biggest challenges or pain points you face in your daily work?** (e.g., repetitive tasks, data analysis, information overload, creative blocks)
- **How could AI tools and technologies potentially address these challenges?** (e.g., automation, pattern identification, personalized recommendations, content generation)
- **Imagine a specific AI tool designed for your job. What would it do?** (e.g., an AI assistant that prioritizes emails, a tool that generates reports from raw data, an AI that helps design presentations)



Refer to **page 17** in the participant guide for this activity.

Debrief time

Share your findings and responses with the whole group.



Embrace AI opportunities

- As a manager, **be curious and creative**, and explore the possibilities and potential of AI for your business.
- **Experiment and innovate with AI solutions** that can enhance your products, services, processes, and customer experiences.
- **Be open and adaptable**, and embrace the changes and challenges that AI brings to your industry and organization.



Key takeaways

- AI technologies are leveraged to drive automation, efficiency, innovation, and strategic advantage.
- Most data and AI projects combine a few methods to extract the full picture.
- The two big components that drive the decision for which method to use are: the **question** you are asking, and the **data** you have.



Summary

Today, we learned about:

- AI for data analytics
- AI implementation

Next session, we will learn about:

- Teams and tools
- Building a data-driven culture
- Ethical and responsible use of data and AI



• End of day 3



DATASOCIETY:

Foundations of Data and AI Literacy For Managers

Day 4



Chat question

“We are all data people!”

What does this phrase imply?



Chat question

Since we are all data people, can we lead all data tasks?



We are all data people

- Whether you realize it or not, every organization uses data to make almost all their decisions.
- However, to truly harness the power of data and AI, and build a successful analytics function, companies need to prioritize establishing the right operational foundation.



Agenda

Day 4

- **Teams and tools**
- Building a data-driven culture
- Ethical and responsible use of data and AI
- What's next?

Data team: Key functions



Data Owner

Typically a senior business leader who is responsible for data domains within the organization.



Data Steward

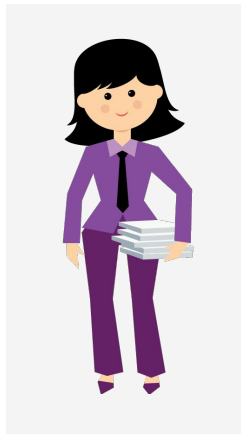
A “go-to” person for data-related questions within a specific data domain, acting as an interface between the technical team and the data owner.



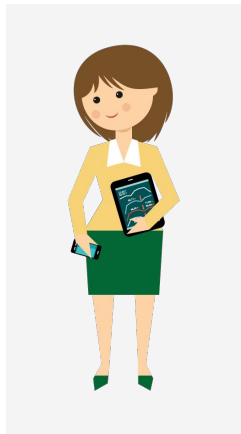
Data Architect

Responsible for designing and maintaining the data infrastructure in line with the policies and standards outlined by the data owner.

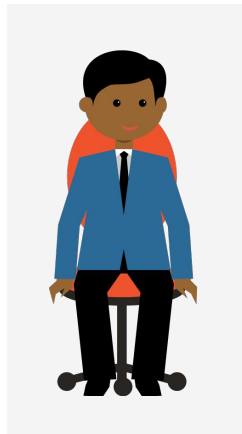
Typical data and AI team personnel



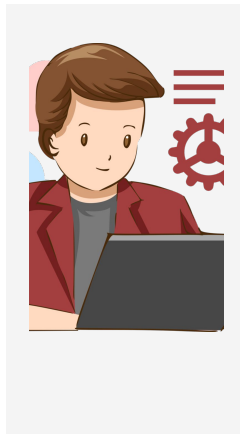
DATA
ANALYST



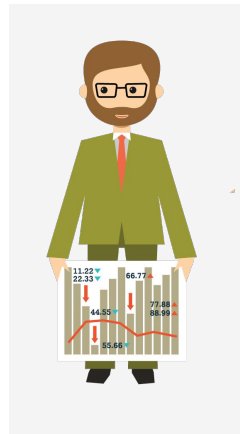
DATA
SCIENTIST



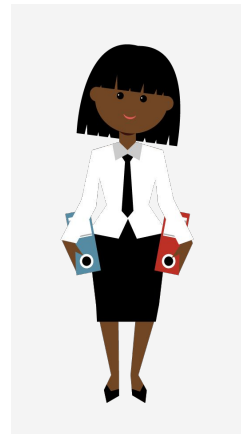
DATA
ENGINEER



SOFTWARE/AI
ENGINEER



MLOPS
EXPERT



DATA SCIENCE
MANAGER

- **Note:** Collaboration is key in data teams, where responsibilities can overlap. Similarly, job titles are not uniform and may differ based on an organization's specific structure.

Data analyst

- Ensures that collected data is relevant and exhaustive while also interpreting the analytics results
- Main responsibilities include:
 - Wrangling the data
 - Managing the data
 - Creating basic analyses and visualizations



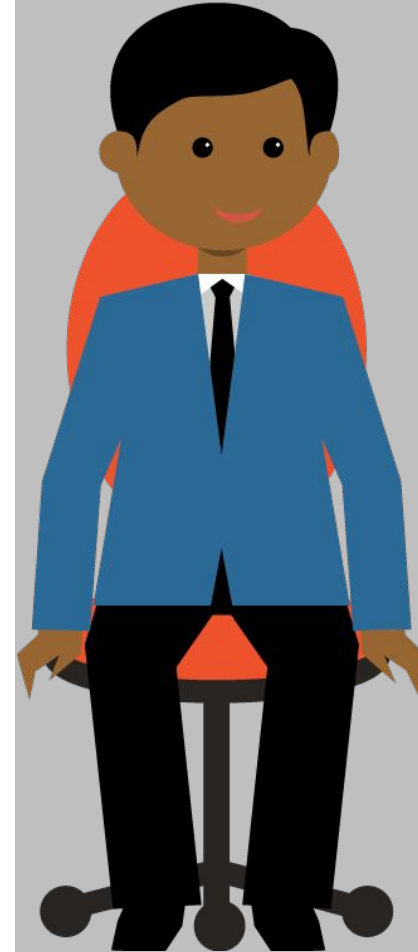
Data scientist

- Builds upon the analysts' data work to develop predictive models and complex algorithms
- Main responsibilities include:
 - Asking the right questions from the data
 - Building more complex predictive models
 - Interpreting the results critically and communicating them well



Data engineer

- Develops the infrastructure to house the data and maintains the structural components
- Main responsibilities:
 - Ensuring data integrity across different data sources
 - Building out additional data warehouses as needed
 - Maintaining data pipelines and access



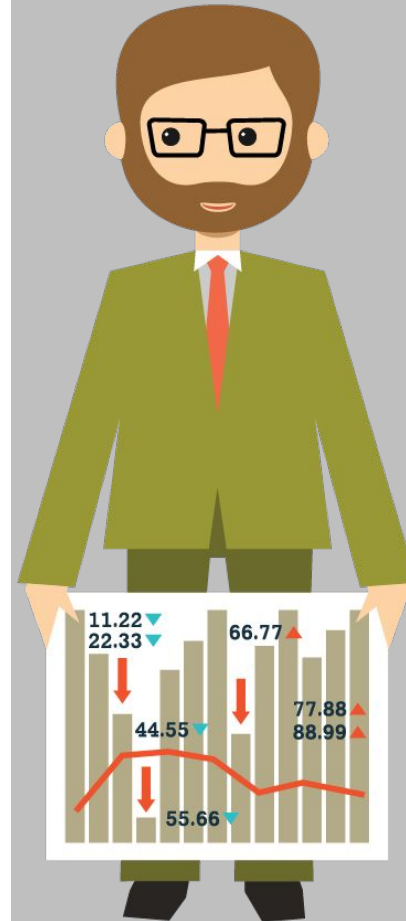
Software/AI engineer

- Responsible for building and maintaining AI software
- Main responsibilities:
 - Construct frameworks that enable AI functions
 - Ensures seamless installation and integration of AI models into software applications
 - Designs front-end systems and integrates data sources



Machine Learning Operations Engineer

- Aims to deploy and maintain machine learning systems in production reliably and efficiently
- Main responsibilities:
 - Requirements engineering
 - System design
 - Implementation and testing
 - Maintenance, support, troubleshooting, etc.



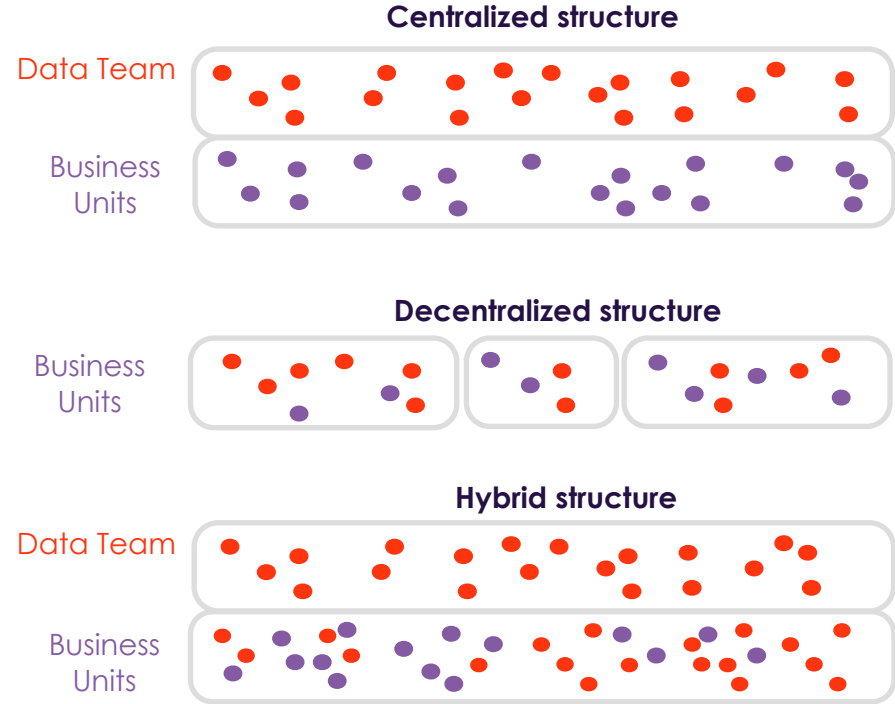
Data science manager

- Oversees and directs data teams and projects and bridges data and non-data people.
- Main responsibilities include:
 - Planning out people and resources for projects
 - Communicating results to executives and stakeholders
 - Running the data science teams



Team structures

- Data analysts
- Business analysts

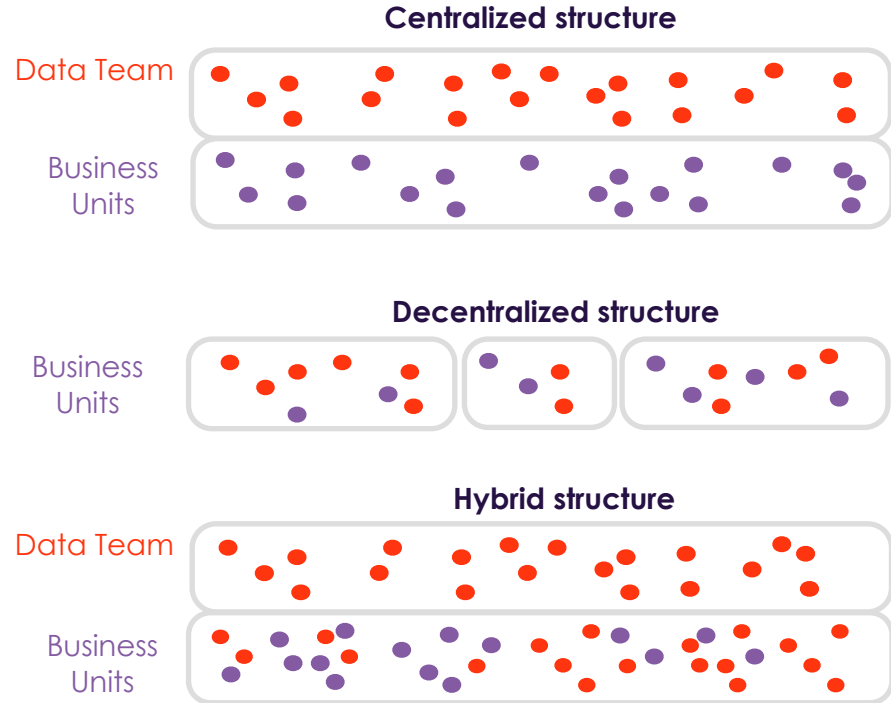




Polling question

Which best describes the structure of the teams in your organization?

- a. Centralized
- b. Decentralized
- c. Hybrid

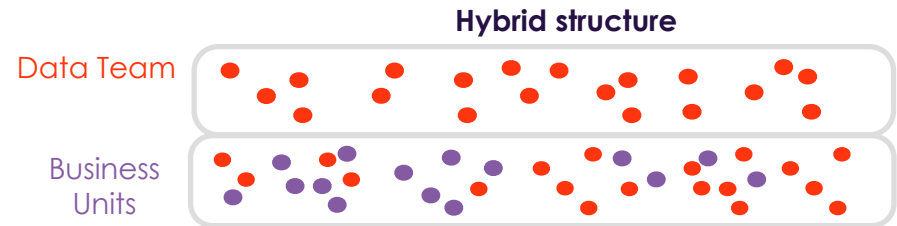
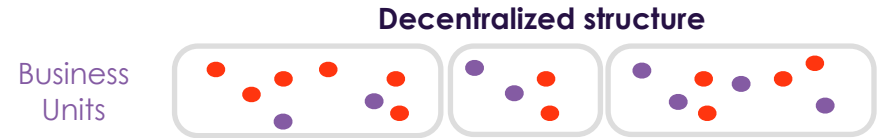
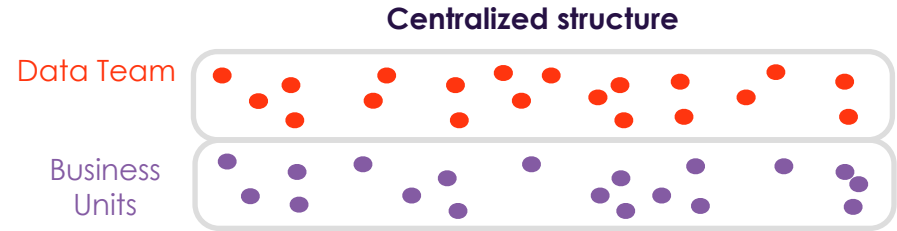


Team structures: Pros and cons

- ✓ easier to standardize team processes
- ✗ harder to coordinate projects to meet strategic goals

- ✓ easier to coordinate projects to meet strategic goals
- ✗ leads to inconsistent & redundant data usage across organization

- ✓ easier to standardize team processes
- ✓ easier to coordinate projects to meet strategic goals



Tools

- There's no shortage of tools in the data analytics space.
- There are different tools for different functions, but most overlap in their offerings.



Tools (cont'd)

A wide range of data tools are available, designed to accelerate data processing and often include built-in AI functionalities. Some common tool include:



Storage

- Databases
- Data warehouses
- Data lakes
- Cloud computing



Cleaning

- Drake
- OpenRefine
- Data Wrangler
- Data Cleaner



Analysis

- Excel
- R
- Python



Visualization

- Excel
- Power BI
- Tableau
- R & RStudio
- Python



Collaboration

- Git
- GitHub

Chat question

When considering the introduction of a new tool, what key factors should you prioritize?

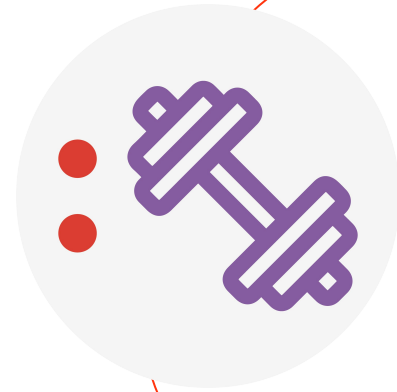


Questions to guide tool selection

- ✓ What specific tasks or problems does the tool need to address?
- ✓ How do the different tools and technologies compare in their functionality, strengths, and weaknesses?
- ✓ Do you have staff who can be trained or know how to use particular tools?
- ✓ How intuitive is the user interface?
- ✓ How well does the tool handle large volumes of data (*or complex workflows*)?
- ✓ Do you have budget constraints you need to be mindful of?
- ✓ Is it on the approved software list?

Exercise: Data tasks and tools reflections

- Take 10 minutes to reflect on the various types of tools and personnel involved in a data project and log your thoughts based on the questions in the participant guide.
- Reflect on your organization's data infrastructure when documenting your thoughts.



Refer to **page 19** in the participant guide for this activity.

Debrief time

Share your findings and responses with the whole group.



Agenda

Day 4

- Teams and tools
- **Building a data-driven culture**
- Ethical and responsible use of data and AI
- What's next?

Chat question

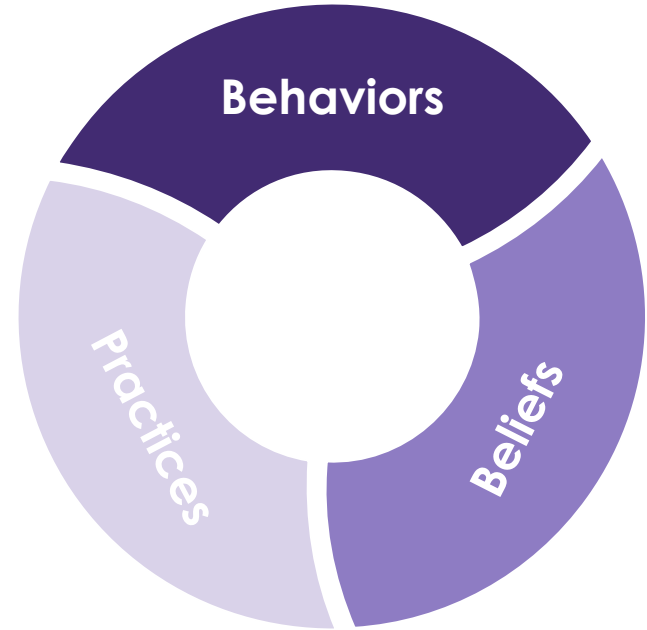
Have you ever wondered:

- What if I have folks new to data on my team?
- How can I model data-driven decision-making?



What does it mean to be data-driven?

- **Data-driven culture** is defined as the behaviors, beliefs, and practices around data to make informed and data-driven decisions.
- To foster a data-driven culture, organizations should:
 - Integrate data into every aspect of the work
 - Mine data for actionable insights
 - Drive change, innovate new products and attract new business



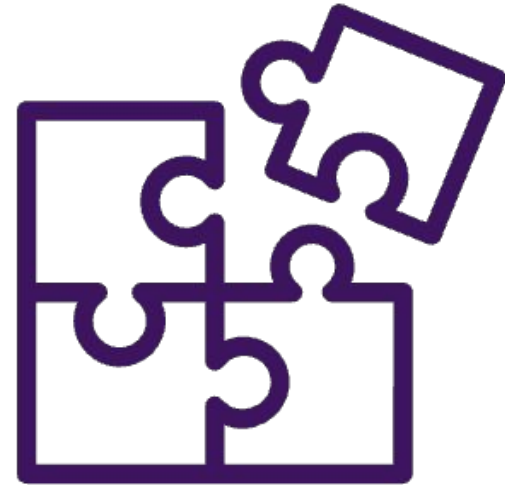
Why is it important to be data-driven?

- **Identify trends.** Trends can inform effective practices, help you become aware of issues, and illuminate possible innovations or solutions.
- **Reduce bias.** Making decisions based on data is far more reliable than ones based on instinct, assumptions, or perceptions.
- **Benchmark performance.** Benchmarking allows staff to connect their actions to business results, which will reveal new opportunities for improvement.

A study from the MIT Center for Digital Business found that organizations driven most by data-based decision making had 4% higher productivity rates and 6% higher profits.

Building blocks of data-driven culture

- A data-driven culture incorporates data and analysis into its business decisions, systems, and processes.
- It can be separated into two main categories:
 - Data infrastructure
 - Data literacy



Building blocks of data-driven culture (cont'd)

Data Literacy



Leadership



Trust

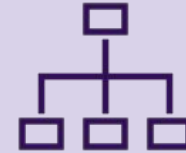


Training

Data Infrastructure



Data
governance



Data
management



Data
quality

How can managers contribute?

To support a data-driven culture, managers must consider the 3 key data governance factors:

People	Processes	Technology
Ensure employees have access to relevant data. Share insights and knowledge. Promote collaborative culture. Empower your team with appreciation and training.	Streamline data collection process. Create data driven guidelines and maintain strong and consistent data governance. Stay informed and keep tabs on new developments in your industry.	Invest in necessary data infrastructure and strategy. Encourage careful and comprehensive data collection methods. Improve and maintain data quality Unified metadata tracking. Automate data management.

How can managers contribute? (cont'd)

- Monitor and help define **data management KPIs** to assess the effectiveness of data and analytics initiatives and identify areas for improvement.
- Ensure that everyone in your team has a **basic understanding** of:
 - Where data comes from (sources)
 - What is the process of collecting data
 - Where data is stored
 - How the data is moved to downstream tools

Key takeaways

- In order to begin cultivating an environment where data can thrive, leaders must understand that:
 - Data is a corporate asset.
 - Data is everyone's business.
 - Everything that can be measured, must be measured.
 - What cannot be measured, cannot be improved.



Break



Agenda

Day 4

- Teams and tools
- Building a data-driven culture
- **Ethical and responsible use of data and AI**
- What's next?

Spotlight: How AI Can Misjudge Potholes



Image source: [Designed by yingyang on Freepik](#)

Data bias

- **Data bias** is inaccuracy or oversight in which some dataset components are more heavily weighted than others.
- A biased dataset can **lead to missed opportunities, skewed outcomes, low accuracy, and analytical errors.**
- Having data representative of the real world is crucial to avoid data bias in machine learning projects.



Data bias example

- A passport application in New Zealand was rejected because the software designed to approve photos thought the man's eyes were closed.
- Even though the applicant submitted a photo with his eyes clearly open, the website's automated system failed to recognize them, rejecting the picture with the message "subject eyes are closed."



**! The photo you want to upload
does not meet our criteria
because:
Subject eyes are closed**



Polling question

- Hiring data scientists is becoming a common strategy for organizations aiming to address ethical data challenges.
- Is this **sufficient** to ensure **responsible use of data**?
 - a. Yes
 - b. No

Data ethics takes collaboration

- Companies often underestimate the complexity of data management and AI, assuming that a few data scientists can handle everything.
- The truth is **data and AI ethics is everyone's domain**, not just the province of data scientists or of legal and compliance teams.
- Data scientists are best positioned to answer questions about experimental design and model development, but they **also** operate best with ethical guardrails in place.



Image source: Created using generative AI tool

What is data ethics?

- Data ethics refers to the **principles and guidelines** that govern how various organizations or bureaus **collect, store, process, analyze and use data**.
- Data ethics must be at the forefront of data initiatives because they help protect individual privacy, build trust with customers, mitigate bias and ensure compliance with laws.



Data ethics at every stage

- It is crucial to ensure data ethics practices are present at every data lifecycle stage.
- Here are the 4 key stages and what should be considered at each stage:

Collecting/Accessing data	<i>Which types or sources of data should be collected, and for what purposes? What type of industry is the data from and for?</i>
Storing the data	<i>What data should be stored and how? Who should be given access to it? What is the level of privacy required (especially for metadata)?</i>
Using/Presenting the data	<i>What cases should be permitted for specific datasets, and how should permissions be managed?</i>
Deleting data	<i>For how long should data be retained, either in live systems or in archive form, and when it should be disposed of?</i>

Chat question



Enova uses AI in its lending system, to provide analytics about finances and credit.



Ocrolus uses AI and machine learning along with human verification to speed up the analysis of financial documents.



Socure uses AI and machine learning to verify identity.

What **potential ethical considerations** should each of these organizations address when implementing AI?



Ethics in AI implementation

Many organizations are increasingly recognizing AI risk evaluation as a central component of their ethical considerations surrounding AI.

- **Legal:** Are we compliant with existing laws?
- **Reputational:** Could the AI system harm our brand or lead to public scrutiny?
- **Operational:** Could the AI system cause harm or malfunction in a way that affects users or operations?
- **Strategic:** Does deploying the technology align with our long-term values and goals?
- **Ethical:** Is the AI reliable, transparent, accountable and fair?



Responsible use of AI

- **Responsible AI** refers to the development, deployment, and use of AI systems that are **ethical, transparent, and accountable**.
- Employees must understand the benefits as well as risks of using AI, and commit to using it ethically.
- To be safe when using AI at work:
 - ✓ Follow your company's policies for using AI.
 - ✓ Use only company-authorized AI services.
 - ✓ Make sure you handle data input into AI systems with caution.
 - ✓ AI can make mistakes! So, always double-check AI-generated results.



Examples of existing data ethics regulations

- National and international governments draft, publish, and enforce data ethics rules.
- Some real-world examples include:
 - European Union's General Data Protection Regulation (GDPR)
 - Health Insurance Portability and Accountability Act (HIPAA)
 - Family Educational Rights and Privacy Act (FERPA)



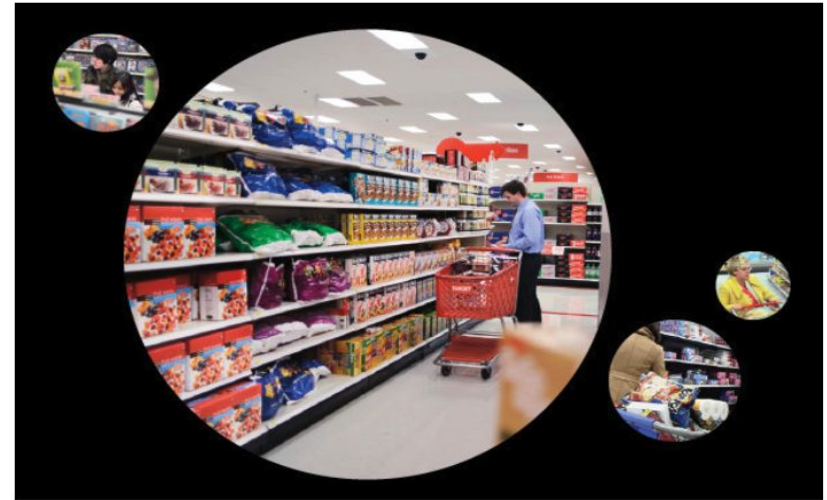
FERPA
Family Educational
Rights & Privacy Act

Predicting pregnancy

- In 2002, Target implemented data analytics to analyze buying patterns in customers.
- New parents often get bombarded with advertising offers, so Target wanted a way to anticipate who is expecting in order to get ahead of the competition.
- They were able to predict pregnancy of their customers based upon their purchases and sent out targeted coupons.

How Companies Learn Your Secrets

By CHARLES DUHIGG FEB. 16, 2012



Antonio Bolfo/Reportage for The New York Times

Source: [NYTimes](#)

Chat question

What potential ethical concerns arose from Target's use of data analytics to identify pregnant customers?



Agenda

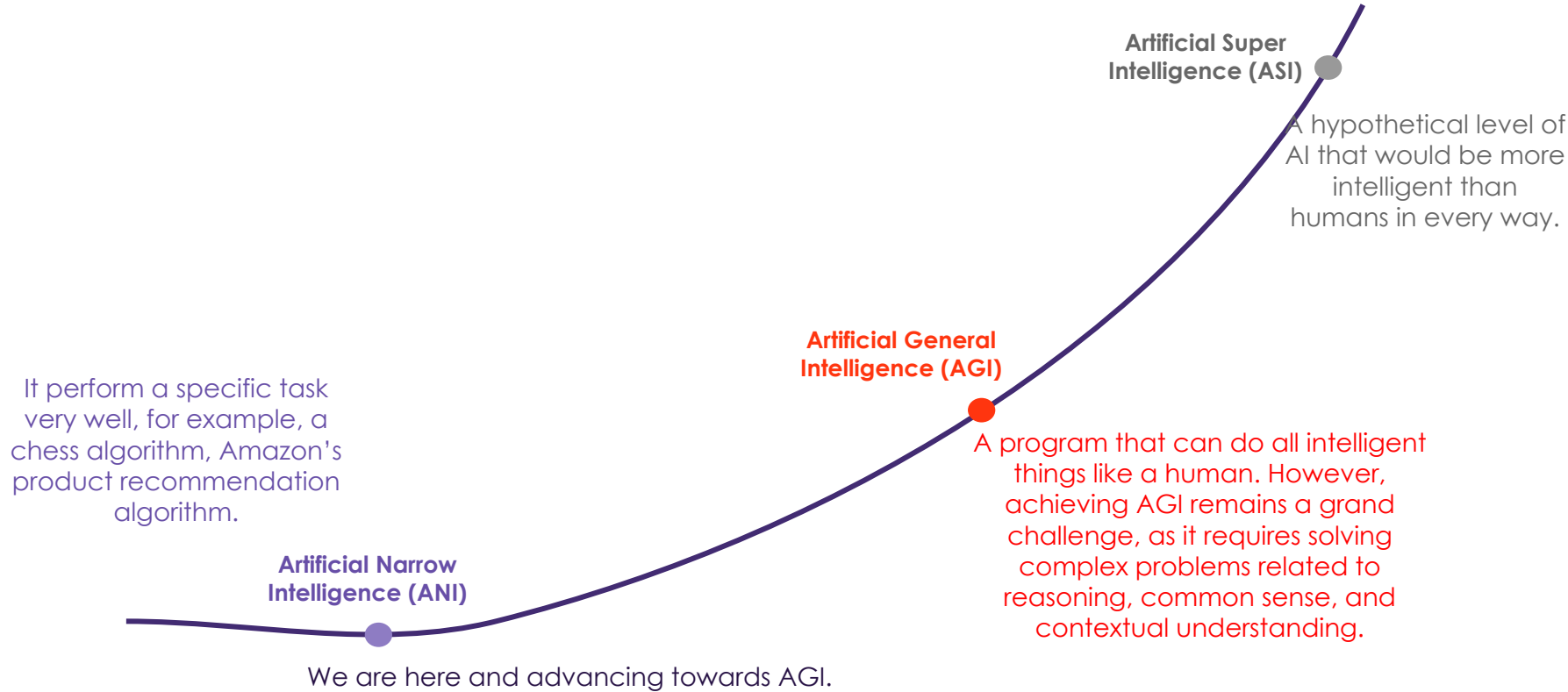
Day 4

- Teams and tools
- Building a data-driven culture
- Ethical and responsible use of data and AI
- **What's next?**

AI's impact

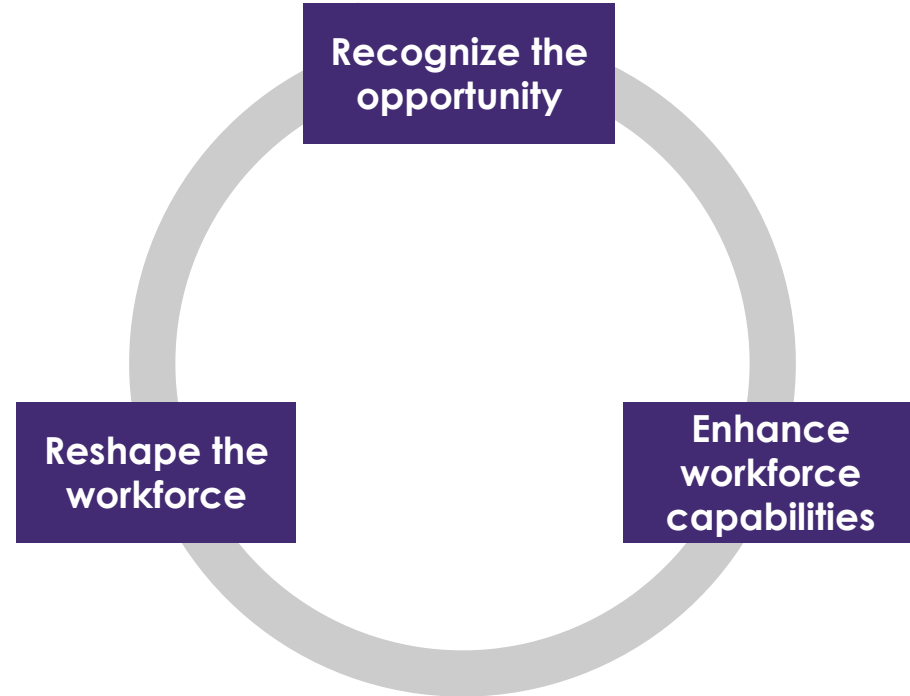
- Every industry is being revolutionized by AI and it is expected to impact many more aspects of operations in all sectors.
 - **Improved business automation:**
Organizations are integrating AI to improve efficacy, profitability and, most immediately, efficiency.
 - **Increased global AI policies and regulations:**
Governments and organizations are prioritizing the establishment of guidelines and policies to promote responsible AI development and utilization.
 - **Economic impact**
Organizations are seriously considering data and AI investments.

The future of AI



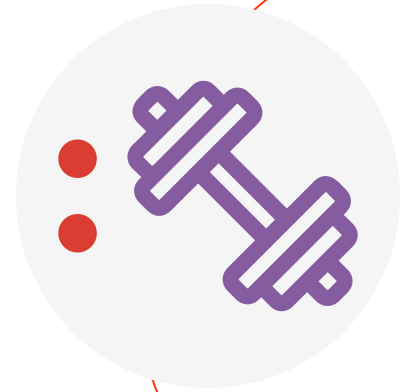
AI and workforce transformation

- Big technological changes don't destroy jobs; they **transform** them.
- Even when they automate existing tasks, they create new, more complex roles and drive economic growth.
- Organizations are trying to understand:
 - What new possibilities does this technology unlock?
 - How will this technology optimize human labor, allowing for greater productivity?



Exercise: Data and AI goals

- Take 10 minutes to reflect on everything we have explored in this course.
- Then, using the roadmap template in the participant guide, outline a plan with specific actions, resources, and timelines for your:
 - Short-term goals (next 3 months)
 - Mid-term goals (6-12 months)
 - Long-term goals (beyond 1 year)



Refer to **page 21** in the participant guide for this activity.

Debrief time

Share your findings and responses with the whole group.



How do you grow further?



Collaboration



Culture



Upskilling



Staying informed



Collaboration

- Volunteer for projects that involve data analysis or visualization, even if it's outside your usual responsibility.
- Foster proactive communication by establishing a shared vocabulary among business teams, data and analytics professionals, and IT departments.



Culture

- Champion data literacy, empower yourself and others with the knowledge and confidence to navigate the complexities of data.
- Enable productive discussions and responsible decision-making around critical issues like data privacy, AI ethics, data bias, and data sharing.



Upskilling

- The rise of AI, automation, and digital ecosystems is transforming the workplace, making retraining and upskilling essential.
- Equip your team with practical data skills. Encourage them to dive into spreadsheet software like Excel or Google Sheets and master the basics of data organization, filtering, and calculations.



Staying informed

- Participate in events that focus on data analytics, AI, or specific industries' use of data to broaden your understanding.
- Take advantage of the talent development programs offered to cultivate a comprehensive and robust skillset.
- Expand your AI knowledge by connecting with communities, groups, and newsletters such as:
 - [Hugging Face community \(linked\)](#)
 - [Reddit AI communities \(linked\)](#)

Conclusion

“Just as the internet has lowered the cost of information, AI will lower the cost of cognition. AI won’t replace humans but humans with AI will replace humans without AI.”

Source: [Professor Karim Lakhani in Harvard Business Review](#)

Summary

Today, we learned about:

- Teams and tools
- Building a data-driven culture
- Ethical and responsible use of data and AI



● End of the course

Questions? Comments?

